

Land Cover Change and Land Surface Temperature Dynamics in Jaipur: A Decadal Analysis (2014-2024)

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1. Introduction and Problem Statement

Urbanization is among the most transformative processes of the 21st century, reshaping landscapes, ecosystems, and climate at both local and regional scales. In rapidly expanding cities, the replacement of vegetated surfaces with impervious materials such as concrete and asphalt disrupts natural land-atmosphere energy exchanges. This leads to a phenomenon known as the **Urban Heat Island (UHI)** effect, where urban areas experience significantly higher temperatures than their rural surroundings.

The consequences of UHI include increased energy demand, deteriorated air quality, reduced human comfort, and exacerbation of health risks—especially during heatwaves. Urban planning and environmental management increasingly rely on understanding UHI dynamics to implement effective mitigation strategies, such as expanding green cover and improving surface reflectivity.

Among India's growing cities, **Jaipur** offers a compelling case study. Located in the semi-arid region of Rajasthan, Jaipur has witnessed substantial land cover transformation over the past decade. Its proximity to natural water bodies like **Man Sagar Lake**, as well as expanding built-up areas, creates a complex thermal landscape influenced by both anthropogenic and ecological drivers.

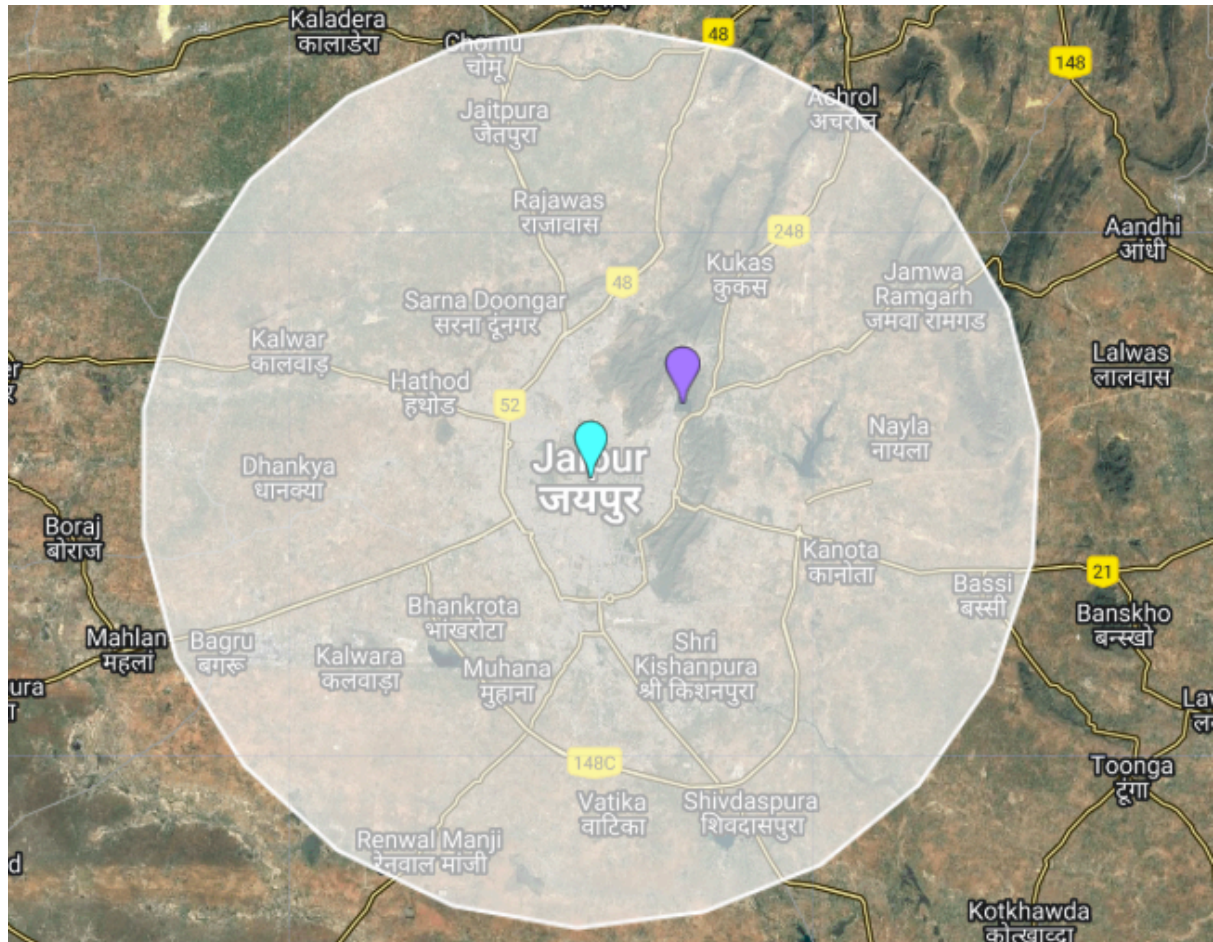
Problem Statement

This study seeks to analyze the relationship between **land cover changes and Land Surface Temperature (LST)** in Jaipur across the years **2014, 2019, and 2024**, using satellite imagery and machine learning-based classification. Specifically, we aim to:

- Classify the area of interest into urban, vegetation, water, and other land cover categories using Indices & MLC. Evaluate the expansion of urban areas using both **sector-based analysis** and **concentric buffer zones**.
- Analyze the spatial correlation between LST and land cover composition per sector and buffer ring.
- Compare classification accuracies using **Random Forest** and **Quadratic Discriminant Analysis (QDA)**.

- Assess the influence of **Man Sagar Lake** on local temperature gradients.

By comparing (totally & individually) land cover classification, LST analysis, and spatial correlation methods, the study aims to provide insights into how urban growth patterns influence thermal dynamics and what role natural elements (like lakes and vegetation) play in moderating heat stress.



2. Data and Its Characteristics

This study relies primarily on multispectral satellite imagery and vector-based ancillary data to perform land cover classification and Land Surface Temperature (LST) analysis for the city of Jaipur. All spatial datasets were acquired from authoritative and openly available sources and preprocessed in accordance with best practices in remote sensing and geospatial analysis.

2.1 Satellite Imagery

Landsat 8 Collection 2 Level-2 surface reflectance products were used for the years **2014**, **2019**, and **2024**. To ensure consistency, the temporal window was limited to **October to December** of each year, aligning with the post-monsoon period and minimizing seasonal variability. A **cloud cover threshold of 10%** was applied during data filtering to select high-quality scenes.

Due to some cloud cover during this time and occasional missing bands in individual scenes, it was **not feasible to rely on single-month imagery**. Therefore, data from **all three months (October–December)** were aggregated for each year.

➤ **Bands Used:**

- **SR_B1 to SR_B7:** Optical reflectance bands (used for indices and classification)
- **ST_B10:** Thermal infrared band (used for LST calculation)

➤ **Indices Computed:**

- NDVI (Normalized Difference Vegetation Index)
- NDBI (Normalized Difference Built-up Index)
- MNDWI (Modified Normalized Difference Water Index)
- BSI (Bare Soil Index)

2.2 Raster Data (& Vector Data) and Ancillary Information

➤ **City Center Point**

A reference point was defined at coordinates **26.9124° N, 75.7873° E** (central Jaipur), and a **30 km buffer** was created around this point to define the **Region of Interest (ROI)**.

➤ **Sector and Buffer Zone Geometry**

- The study area was divided into **eight 45-degree angular sectors**, radiating from the city center.
- A second spatial framework involved **concentric buffer rings** spaced every 2 km (2 km to 10 km) around **Man Sagar Lake**, allowing for gradient-based LST analysis.

➤ **Training Data for Classification**

Manually digitized **training points** were collected for four land cover classes:

- **Urban / Built-up**
- **Vegetation**
- **Water Bodies**
- **Others** (e.g., barren, sand, rocky)

These were created in Google Earth Engine for each year and exported as GeoJSON files for use in supervised classification.

Table: *Training Data Class Distribution*

Year	Urban	Vegetation	Water	Others	Total
2014	52	51	34	58	195

2019	52	57	63	48	220
2024	58	47	36	69	210

3. Methodology

This study combines index-based classification, machine learning-based classification, and land surface temperature (LST) estimation techniques to analyze spatio-temporal land cover changes and their thermal impact across Jaipur. The entire methodology was implemented using **Google Earth Engine (GEE)** and **Google Colab (Python)**.

3.1 Pre-processing

All satellite images were preprocessed to ensure radiometric consistency and cloud-free surface representation across the temporal scope of the study (2014, 2019, and 2024). The following steps were performed using the Google Earth Engine platform.

3.1.1 Scale Factor Application

Landsat 8 Collection 2 Level-2 products provide data in scaled digital numbers (DN) that require conversion to physical reflectance and temperature values.

- Optical Bands (SR_B1 to SR_B7):**
 Surface reflectance values were derived using the scaling factor provided in USGS documentation:

$$\text{Reflectance} = \text{DN} \times 0.0000275 - 0.2$$
- Thermal Band (ST_B10):**
 The brightness temperature in Kelvin was computed as:

$$\text{Tb} = \text{DN} \times 0.00341802 + 149.0$$
 These operations standardize input data across years and ensure physical comparability.

3.1.2 Cloud Masking and Quality Assessment

To minimize the influence of atmospheric obstructions, a cloud and snow mask was applied using the **QA_PIXEL** quality assurance band.

- Masked Features:**
 - Clouds (bit 3)
 - Cloud Shadows (bit 4)
 - Snow (bit 5)

A bitwise mask operation filtered out these artifacts, resulting in spatially consistent composites.

3.1.3 Image Compositing Approach

For each study year (2014, 2019, 2024), multiple scenes between **October 1st and December 31st** were collected. These were filtered using:

- **Cloud Cover < 10%**
- **Region of Interest (ROI):** A 30 km buffer from Jaipur's city center

The resulting image collections were then composited using the **median reducer**, which selects the median value for each pixel across time. This approach reduces residual cloud contamination and seasonal outliers while preserving landscape characteristics.

3.1.4 Spatial Consistency and Resolution

All processing was conducted at a **30-meter spatial resolution**, in the **WGS 84 / EPSG:4326** coordinate system. This ensures uniformity in classification and spatial analysis.

3.2 Land Cover Classification Approaches

Land cover classification was performed using two approaches: a rule-based method utilizing spectral indices (Section 3.2.1), and a supervised machine learning approach (Section 3.2.2). Both methods were applied to Landsat composites from three time periods: 2014, 2019, and 2024. Each classified the landscape into four thematic classes:

- **Urban/Built-up**
- **Vegetation**
- **Water Bodies**
- **Others (bare soil, rocky, or mixed types)**

3.2.1 Index-Based Classification

The index-based classification approach uses **empirically defined thresholds** applied to remote sensing indices that are sensitive to specific land cover types. The following indices were computed and thresholded within Google Earth Engine.

Index Descriptions and Roles

- **NDVI – Normalized Difference Vegetation Index**
$$\text{NDVI} = \frac{\text{SR_B5} - \text{SR_B4}}{\text{SR_B5} + \text{SR_B4}}$$

Used to identify green vegetation. High values (>0.3) indicate healthy vegetation.

- **NDBI – Normalized Difference Built-up Index**
$$\text{NDBI} = \frac{\text{SR_B6} - \text{SR_B5}}{\text{SR_B6} + \text{SR_B5}}$$

Designed to highlight urban/built-up areas. High NDBI often coincides with impervious surfaces.

- **MNDWI – Modified Normalized Difference Water Index**
$$\text{MNDWI} = \frac{\text{SR_B3} - \text{SR_B6}}{\text{SR_B3} + \text{SR_B6}}$$

Enhances water bodies by suppressing vegetation and built-up noise.

- **BSI – Bare Soil Index**

$$BSI = (SR_B6 + SR_B4) - (SR_B5 + SR_B2) / (SR_B6 + SR_B4) + (SR_B5 + SR_B2)$$

Highlights bare soil or mixed land.

Classification Logic and Thresholds

Each pixel was evaluated based on the following decision rules:

Land Cover Class	Index Conditions
Urban	$NDBI > 0.15$ and $NDVI < 0.5$
Vegetation	$NDVI > 0.37$ and $MNDWI < 0.3$
Water	$MNDWI > 0.01$
Others	$BSI \leq 0.025$

These thresholds were chosen based on literature values and visual tuning to local conditions.

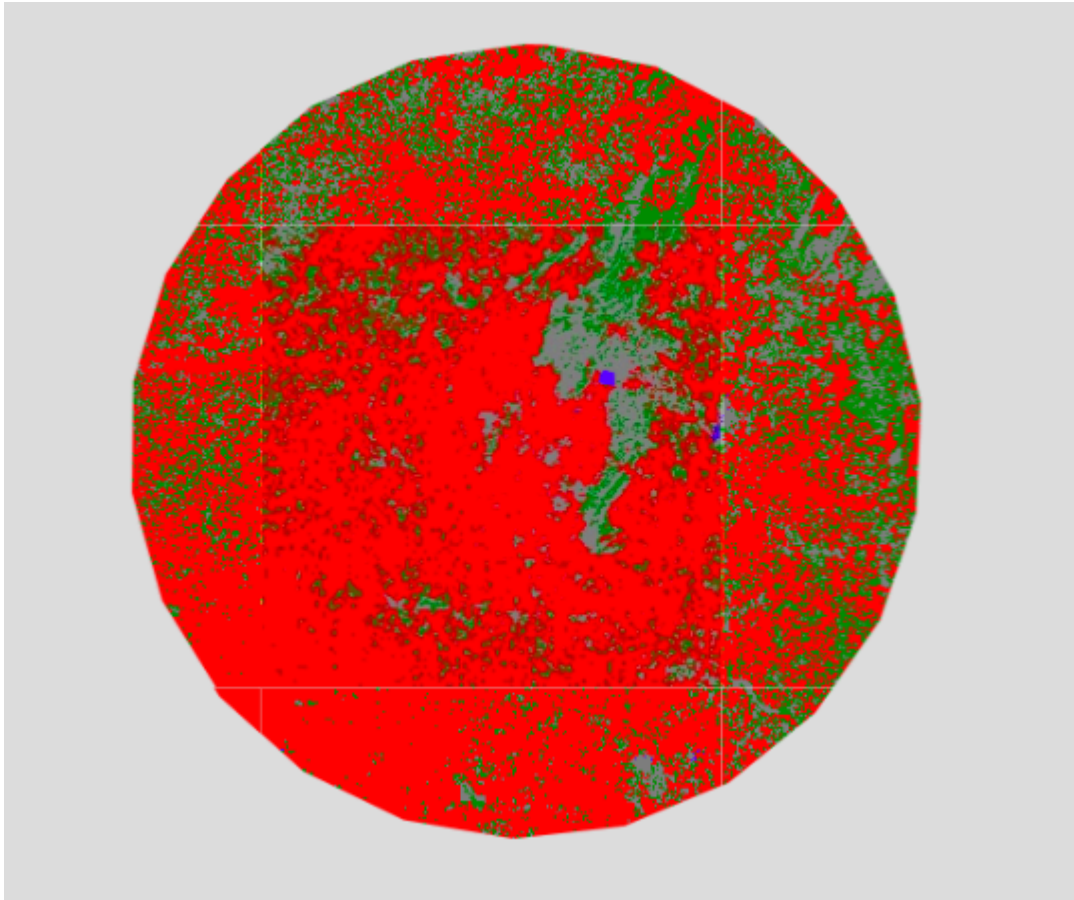
Implementation in Google Earth Engine

- Indices were calculated for each composite image (2014, 2019, 2024).
- A rule-based map algebra approach was used to apply conditional logic.
- Pixels meeting each class's conditions were assigned a unique class code:
 - 0 = Urban
 - 1 = Vegetation
 - 2 = Water
 - 3 = Others

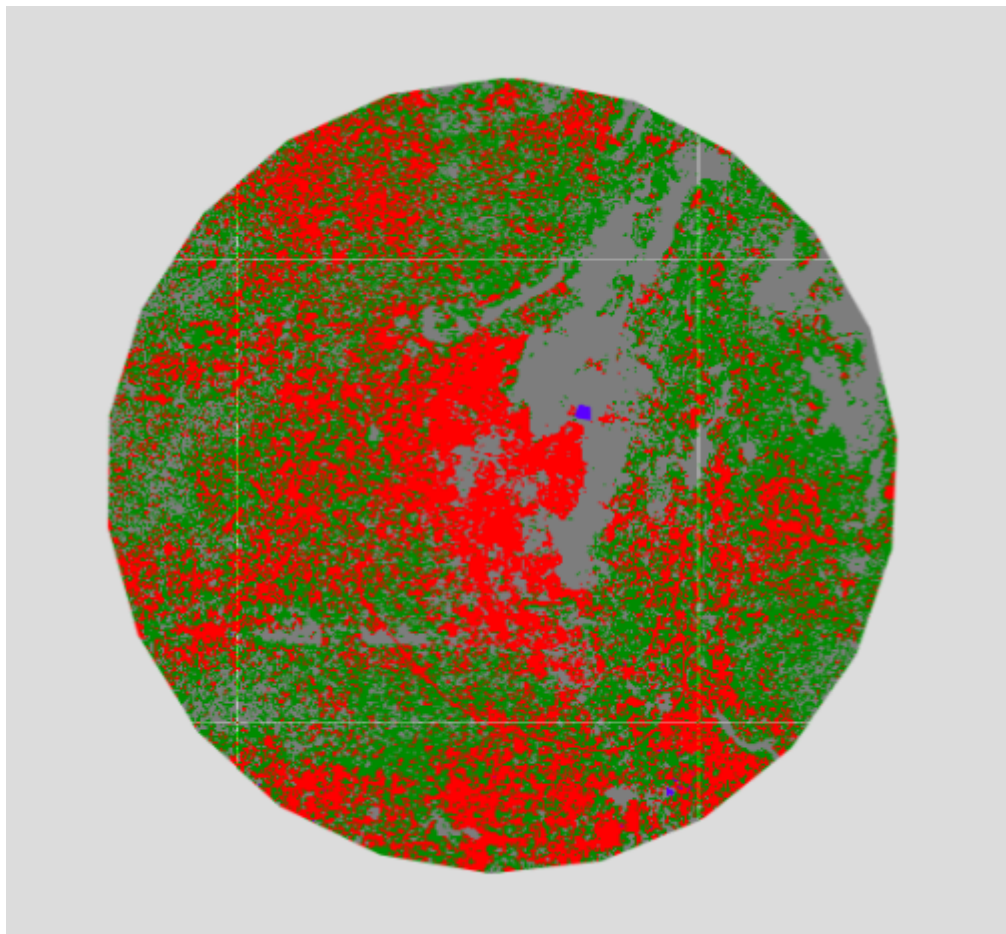
The resulting classification images were visualized using a consistent color palette and used for spatial statistical extraction.

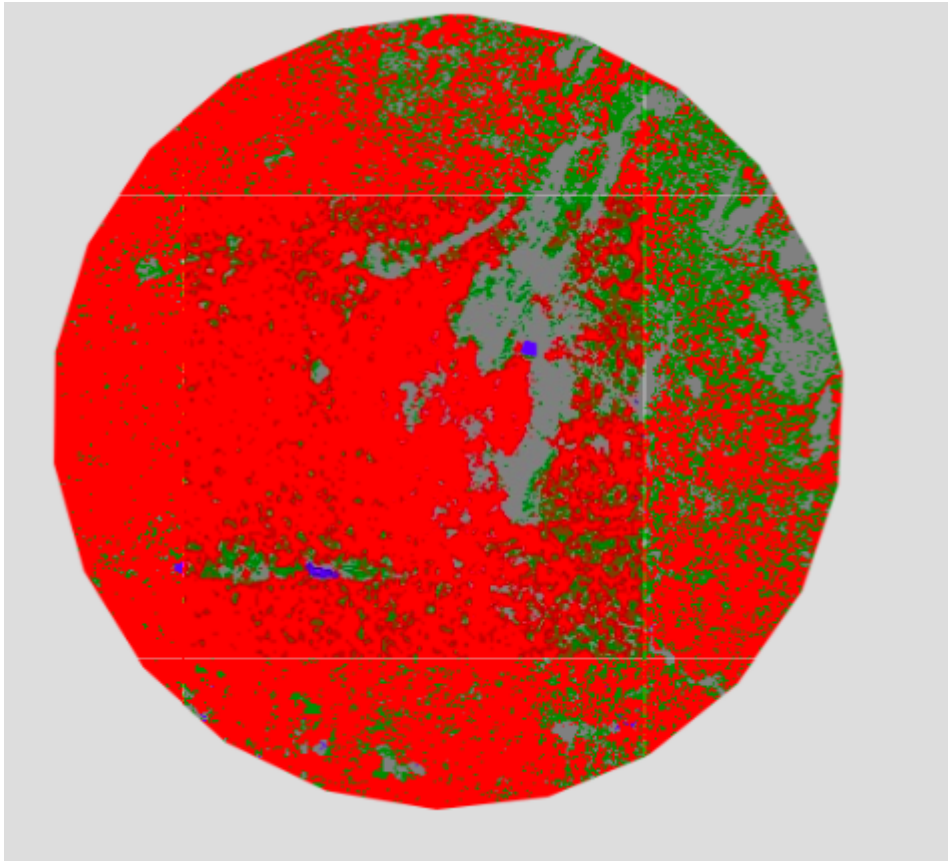
Index Based Classification

2014



2019





3.2.2 Machine Learning Classification

In addition to the index-based rule classification, a supervised machine learning approach was implemented using two algorithms: **Quadratic Discriminant Analysis (QDA)** and **Random Forest (RF)**. These models were trained using manually labeled point features, with input features derived from satellite spectral bands and calculated indices.

Image Stack Creation and Feature Selection

For each year (2014, 2019, 2024), image stacks were created to serve as input features for classification. These stacks combined both raw spectral bands and derived indices.

This resulted in an **11-band feature vector per pixel**. The image stacks were exported from Google Earth Engine in **.tif** format and ingested in Python using **rasterio**.

Model Development

Two supervised classifiers were trained and evaluated:

Quadratic Discriminant Analysis (QDA)

- Assumes multivariate normal distributions with distinct covariance matrices per class.
- Regularization parameter: **reg_param = 0.5**

Random Forest (RF)

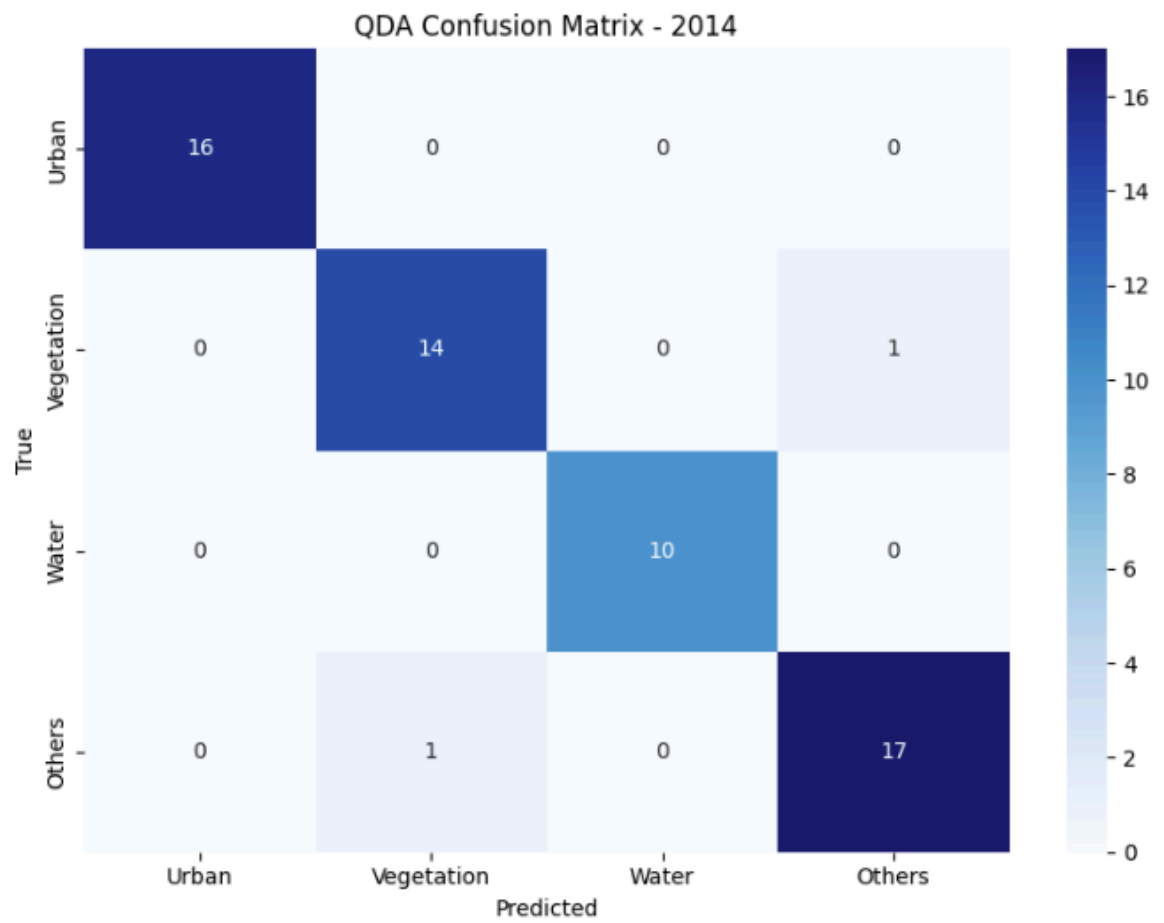
- Ensemble of decision trees using bagging.
- Parameters:
 - `n_estimators` = 200
 - `max_depth` = 10
 - `min_samples_leaf` = 5
 - `random_state` = 42

Accuracy Assessment and Validation

Each model was trained using a **stratified 70/30 train-test split**. Model performance was assessed using:

- **Overall Accuracy**
- **Kappa Coefficient**
- **Mean Squared Error (MSE)**
- **Confusion Matrix**
- **Precision, Recall, F1-score** (per class)

Figure: Confusion Matrix Heatmaps for QDA and RF – 2014
(similar figures included for 2019 and 2024 in the appendix)



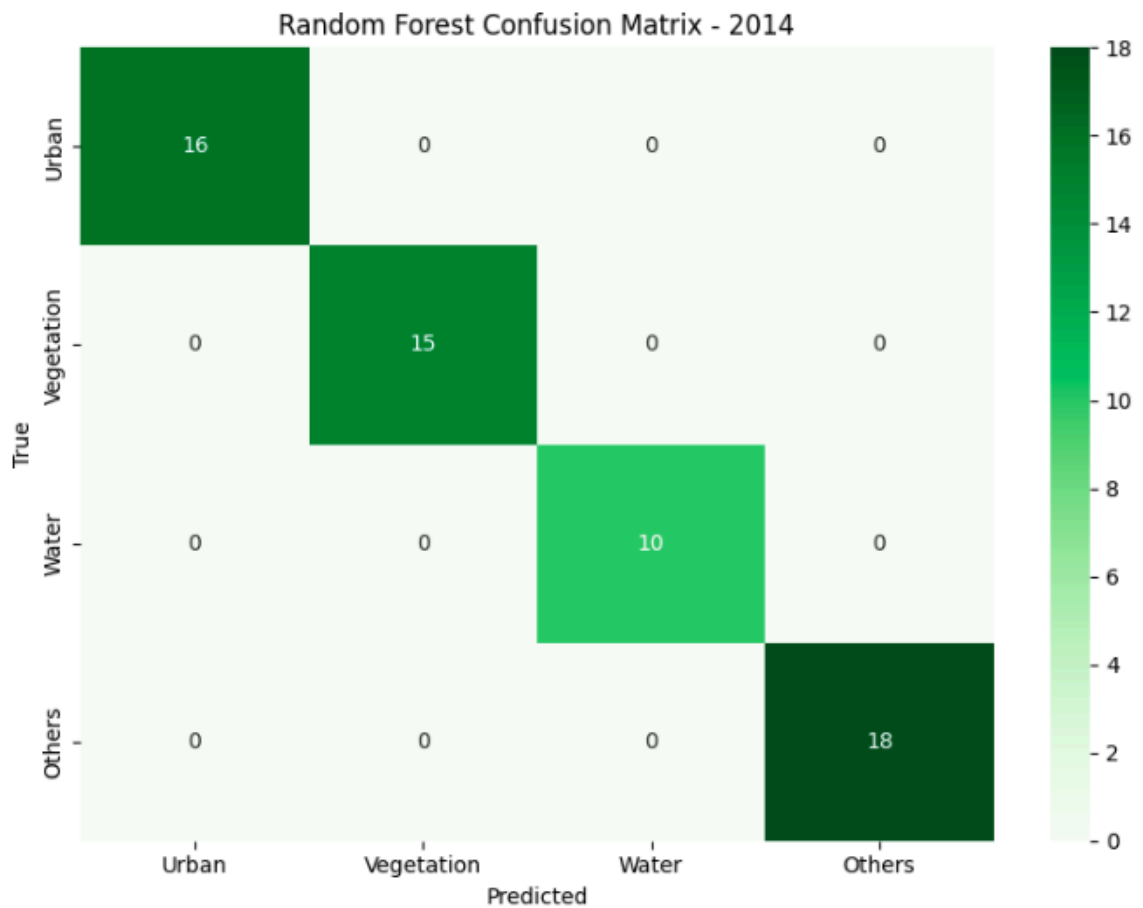


Figure: Bar Plot Comparing Overall Accuracy QDA vs RF (2014, 2019, 2024)

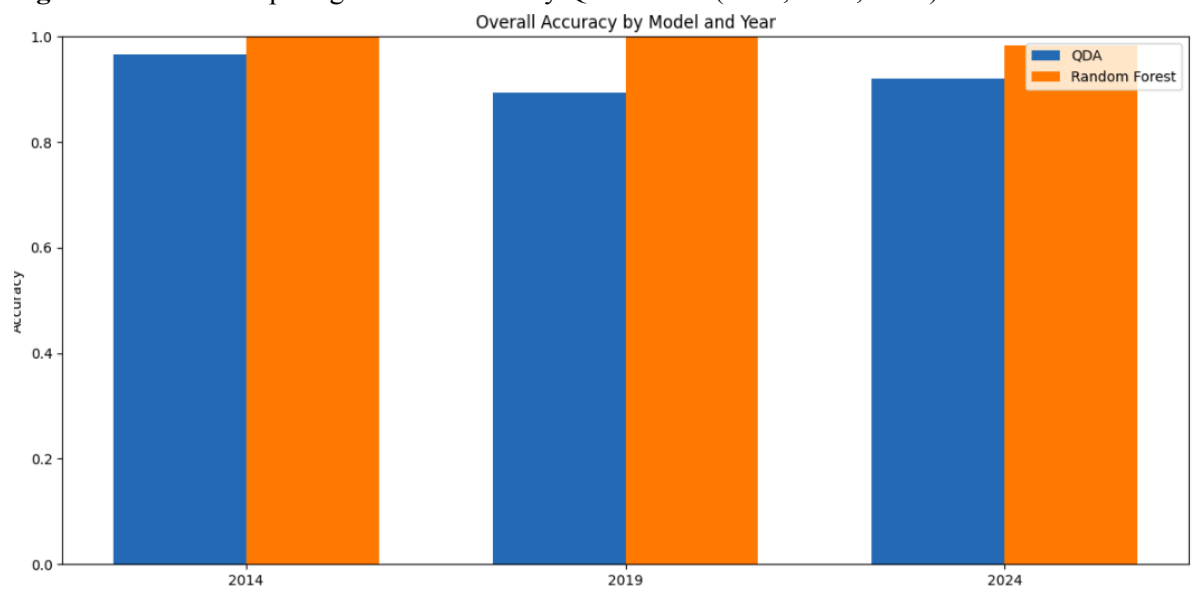


Table A: Model Performance (Overall Accuracy, Kappa, MSE)

Year	Model	Overall Accuracy	Kappa Coefficient	Mean Squared Error
2014	QDA	96.61%	0.9542	0.1356

2014	Random Forest	100.00%	1	0
2019	QDA	89.39%	0.8574	0.7576
2019	Random Forest	100.00%	1	0
2024	QDA	92.06%	0.8915	0.7143
2024	Random Forest	98.41%	0.9785	0.1429

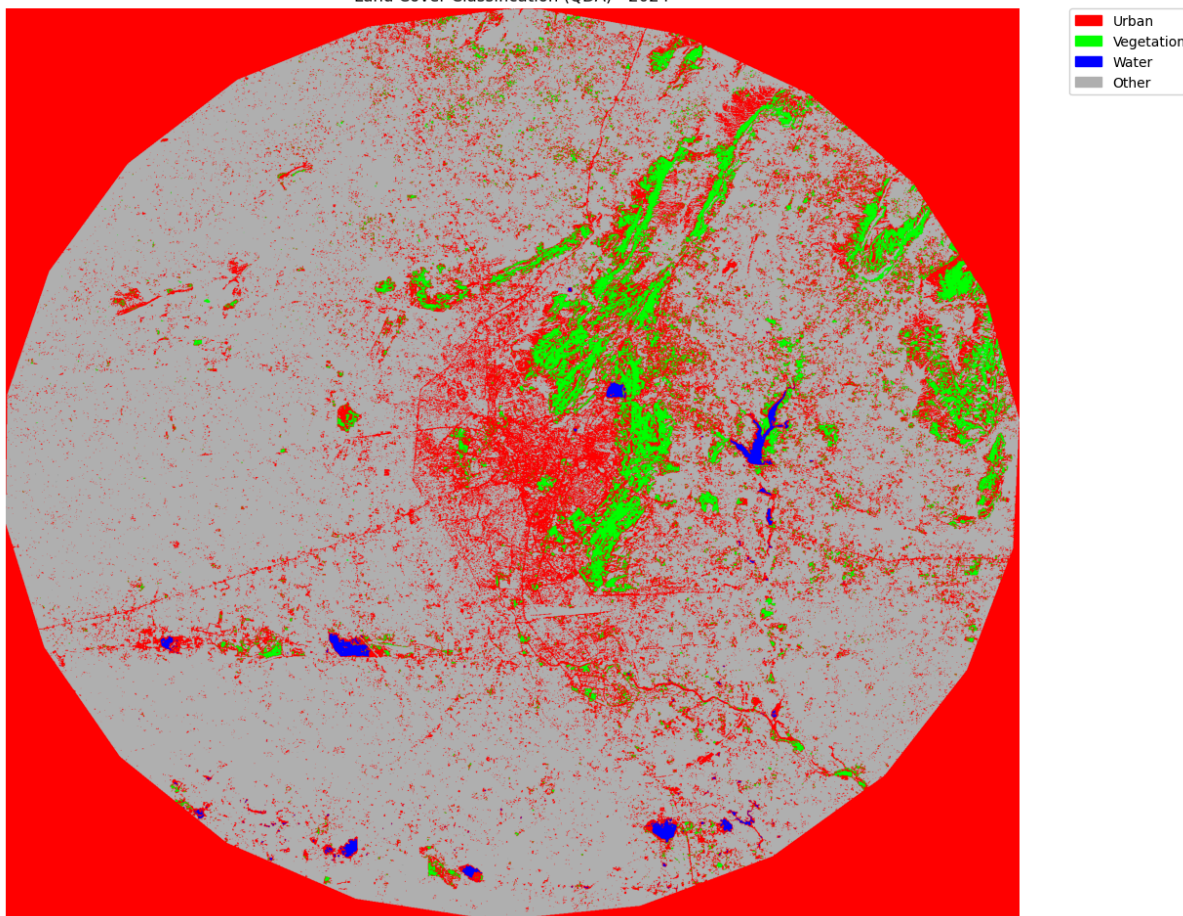
Table B: Precision, Recall, and F1-score by Class and Year

Year	Model	Class	Precision	Recall	F1-score
2014	QDA	Urban	1	1	1
2014	QDA	Vegetation	0.93	0.93	0.93
2014	QDA	Water	1	1	1
2014	QDA	Others	0.94	0.94	0.94
2014	Random Forest	All Classes	1	1	1
2019	QDA	Urban	0.75	0.94	0.83
2019	QDA	Vegetation	0.89	1	0.94
2019	QDA	Water	1	1	1
2019	QDA	Others	1	0.57	0.73
2019	Random Forest	All Classes	1	1	1
2024	QDA	Urban	1	0.71	0.83
2024	QDA	Vegetation	1	1	1
2024	QDA	Water	1	1	1
2024	QDA	Others	0.81	1	0.89
2024	Random Forest	Urban	0.94	1	0.97
2024	Random Forest	Vegetation	1	1	1
2024	Random Forest	Water	1	1	1
2024	Random Forest	Others	1	0.95	0.98

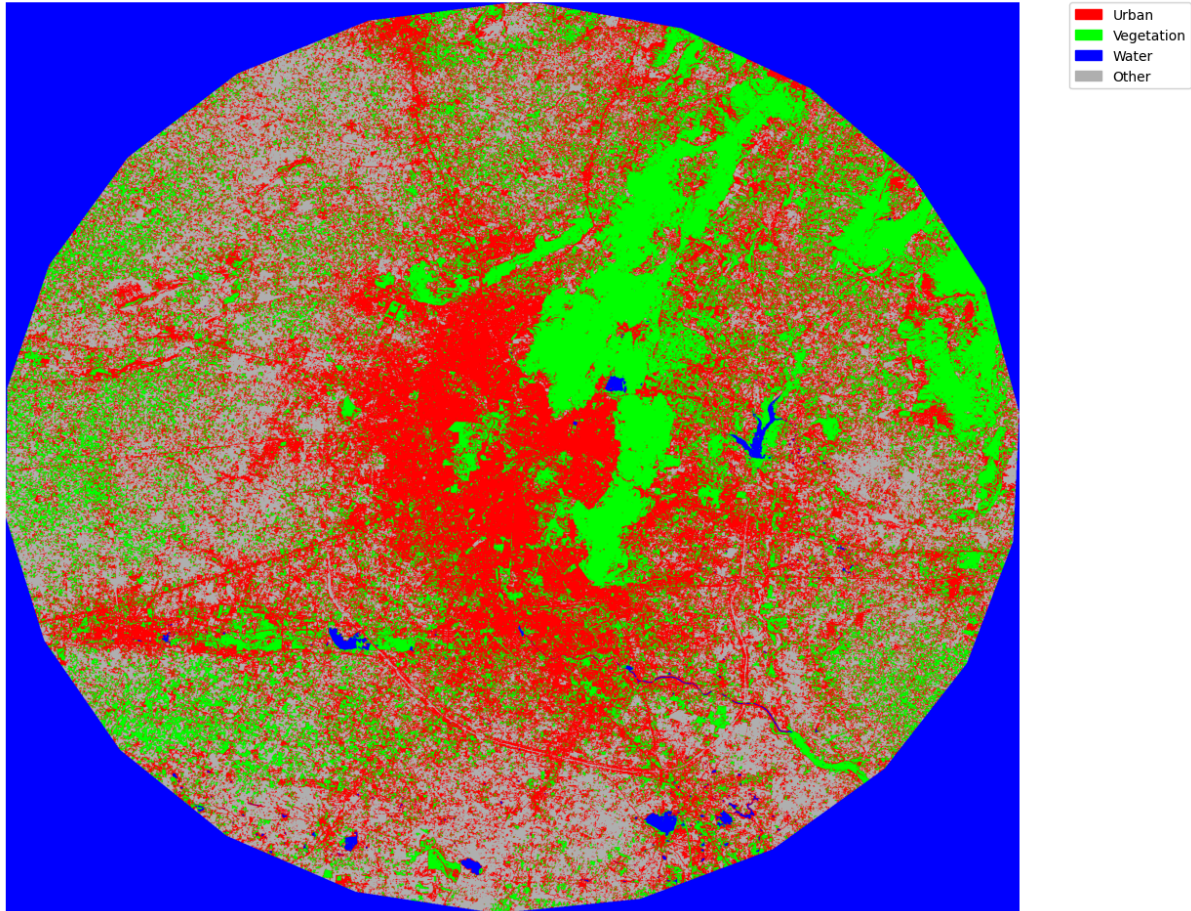
Data Integration and Classification Output

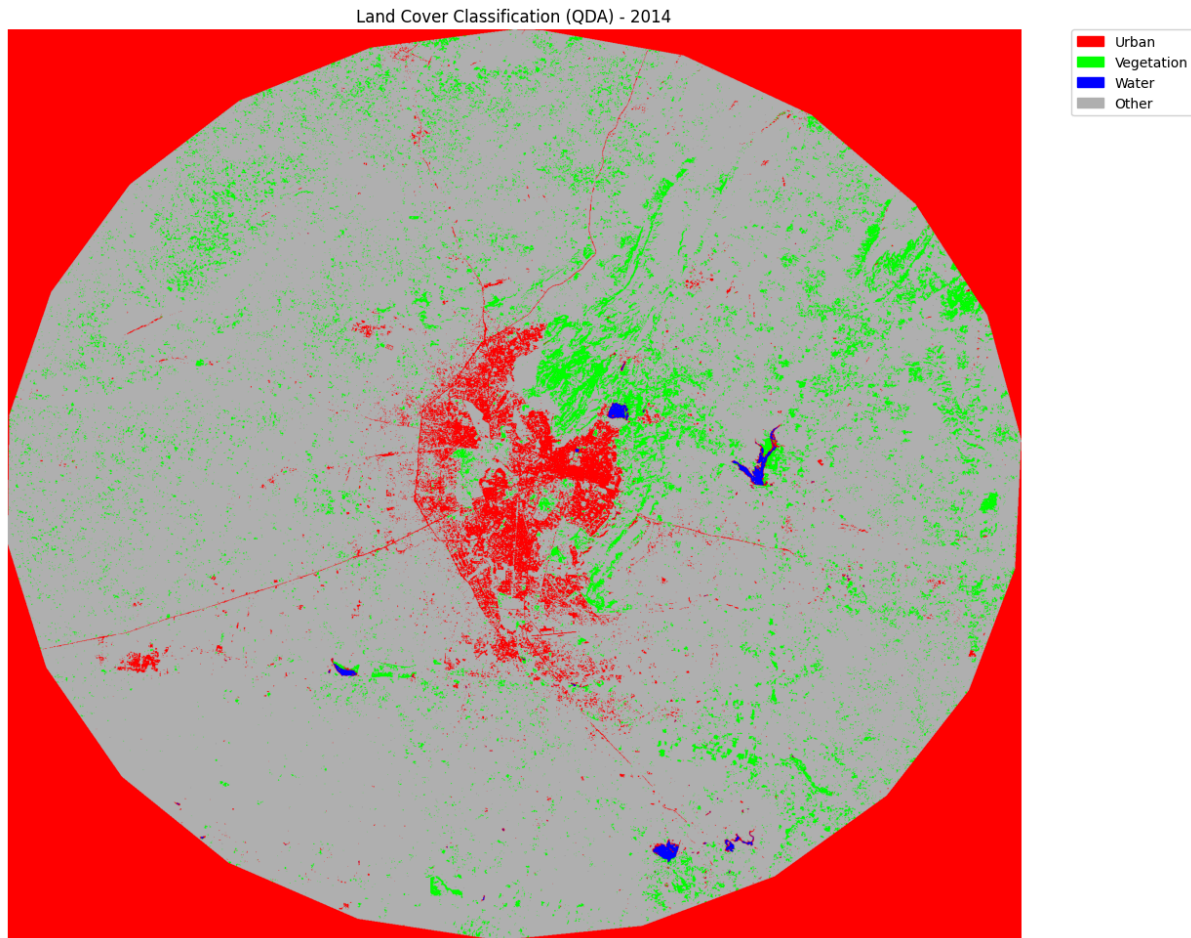
- Final classified images (QDA-based) were exported as GeoTIFFs and colored rasters.
- Classification results were overlaid on base maps to visually inspect accuracy.
- The classified rasters were used in subsequent **zonal and temporal analyses**.

Land Cover Classification (QDA) - 2024



Land Cover Classification (QDA) - 2019





3.3 Land Surface Temperature Derivation

Understanding urban heat dynamics requires accurate estimation of **Land Surface Temperature (LST)**. In this study, the **Mono-Window Algorithm (MWA)** was used to estimate LST from **Landsat 8** thermal data (band 10). This section outlines the preconditions, equations, and implementation used in deriving LST for the years 2014, 2019, and 2024.

3.3.1 Thermal Band Preprocessing

The **ST_B10** band from the Landsat 8 Level-2 Surface Reflectance product was used. Prior to LST calculation, the raw thermal digital numbers were converted to **Top of Atmosphere (TOA) Brightness Temperature** using the scale factor:

$$T_b = ST_B10 \times 0.00341802 + 149.0$$

where:

- T_b : Brightness Temperature in Kelvin

This conversion was applied across all image composites using a function within GEE.

3.3.2 Emissivity Estimation Using NDVI

Since emissivity varies with land cover, **NDVI-based emissivity** was estimated as follows:

1. **NDVI** was computed using bands SR_B5 (NIR) and SR_B4 (Red):
$$\text{NDVI} = \frac{\text{SR_B5} - \text{SR_B4}}{\text{SR_B5} + \text{SR_B4}}$$
2. **Proportion of Vegetation (Pv)** was calculated:
$$\text{Pv} = (\text{NDVI} - 0.2) / (0.5 - 0.2).clamp(0, 1)$$
3. **Land surface emissivity (ϵ)** was then estimated using:
$$\epsilon = 0.004 \times \text{Pv} + 0.986$$

3.3.3 Mono-Window LST Algorithm

The **Mono-Window Algorithm** was then applied:

$$\text{LST} = \text{Tb} / [1 + (\lambda \times \text{Tb} / \rho) \times \ln(\epsilon)]$$

Where:

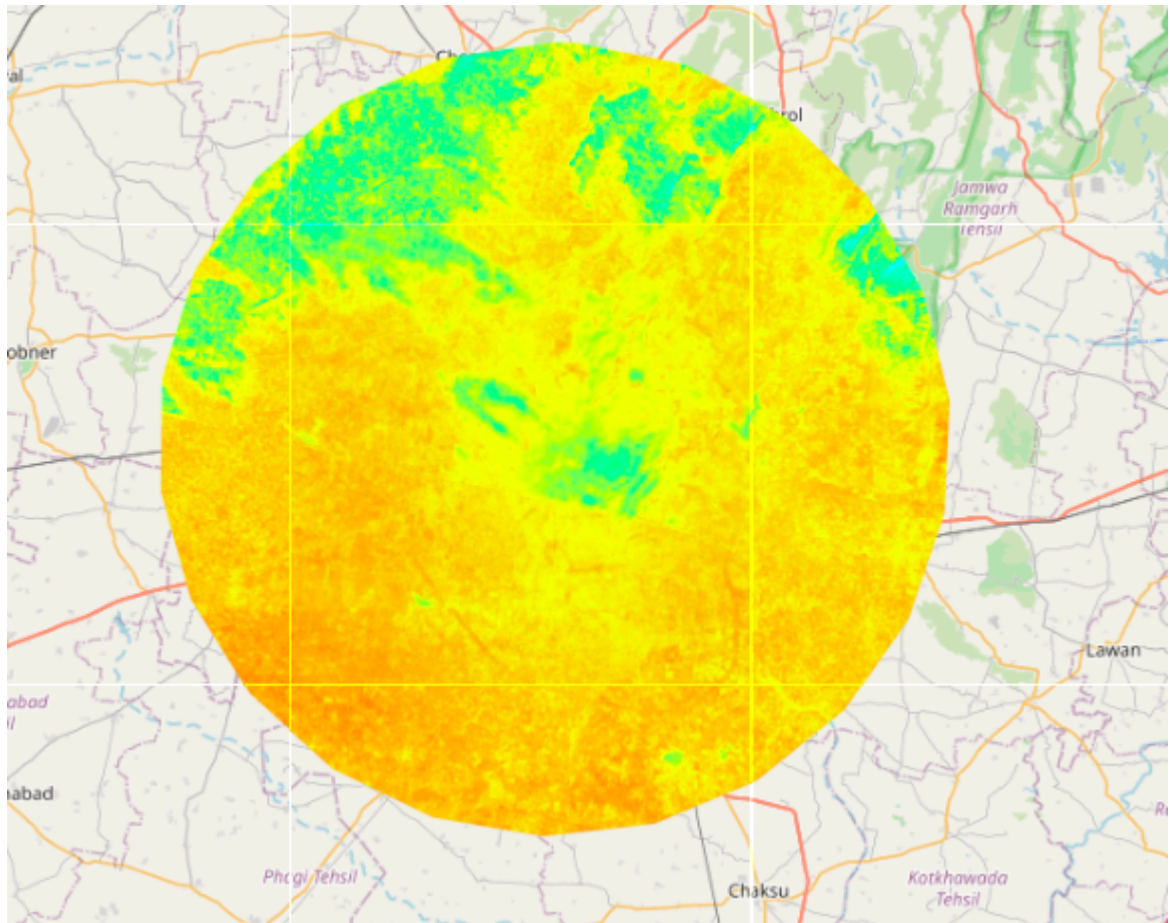
- Lambda (λ) = 10.8×10^{-6} meters — this is the center wavelength for Landsat 8 Thermal Band 10.
- Rho (ρ) = 14,380 — a constant derived from Planck's Law.
- Tb = Brightness temperature (obtained from the thermal band of the satellite image).
- Epsilon (ϵ) = Surface emissivity, which represents how efficiently the surface emits thermal radiation.

The resulting **LST values were in Kelvin**, which were later converted to **Celsius** by subtracting 273.15.

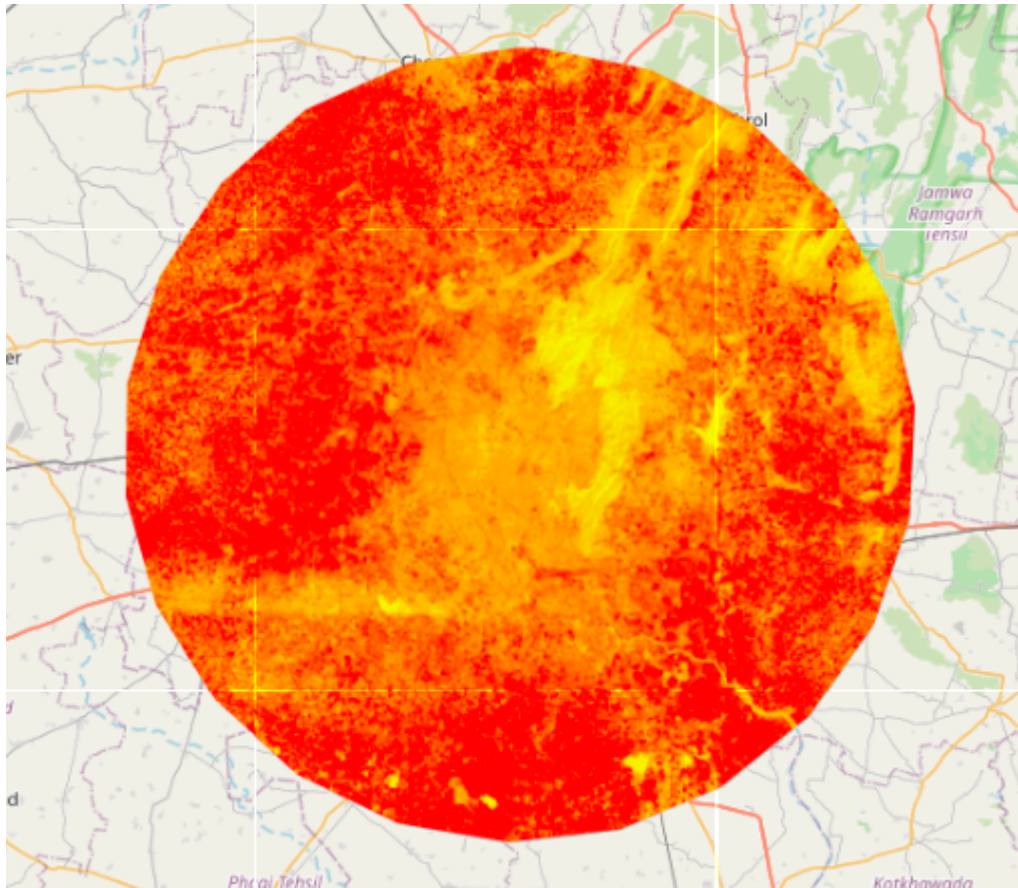
3.3.4 Implementation and Visualization

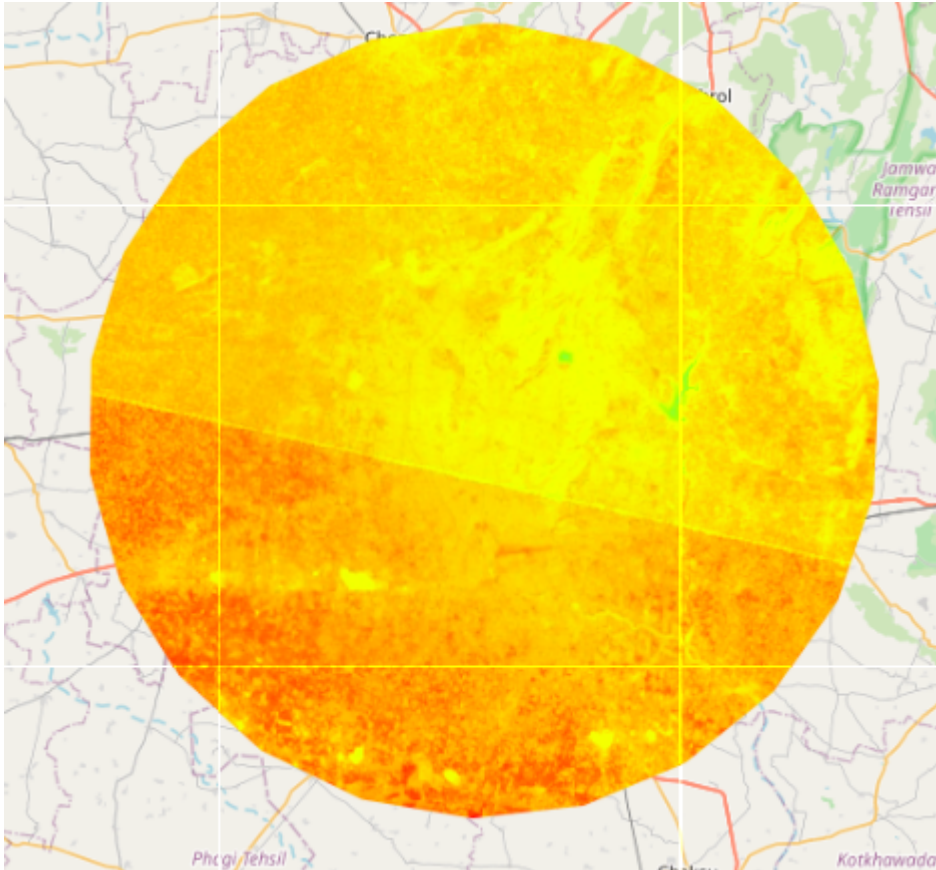
- This process was automated in Google Earth Engine using `image.expression()` functions.
- Three LST rasters (2014, 2019, 2024) were created and visualized using a consistent color palette.

Figure: *Land Surface Temperature (LST) Maps – 2014, 2019, 2024*
2014



2019





These LST layers were then used in both the **sector-wise analysis** and **buffer analysis around Man Sagar Lake**, as detailed in the following spatial analysis section.

3.4 Spatial Analysis Methods

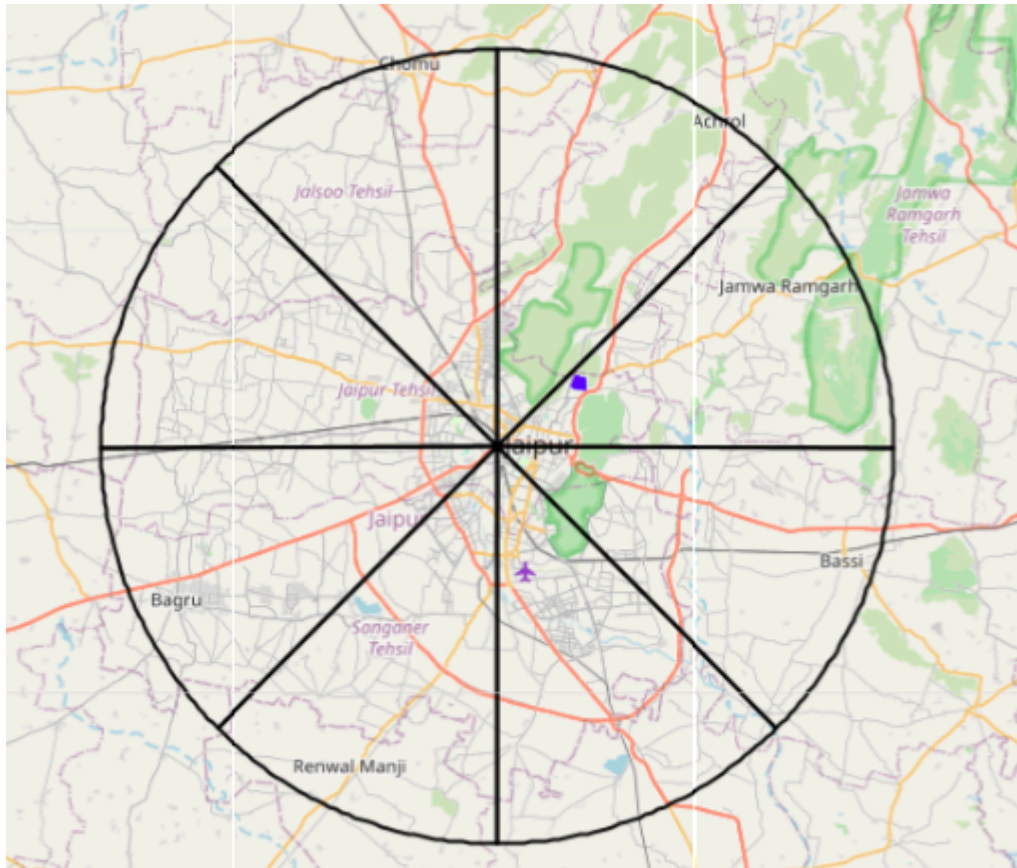
To understand the spatial dynamics of land cover change and thermal behavior, we employed two targeted spatial strategies: (1) **sector-wise radial analysis from the city center**, and (2) **buffer-based analysis around a major water body**. These methods helped quantify spatiotemporal patterns of urban expansion, vegetation loss, and their impact on land surface temperature (LST).

3.4.1 Sector Analysis

a) Sector Creation

- The region of interest (ROI) — a 30 km radius around Jaipur city center — was divided into **eight equal 45° sectors** radiating outward.
- Each sector polygon was generated using geographic trigonometry and `ee.Geometry.Polygon` functions in Google Earth Engine.

Figure: Map showing 8 sectors around city center (45° each)



b) Land Cover Statistics per Sector

For each sector and each year (2014, 2019, 2024), the classified images (both index-based and QDA) were analyzed to calculate:

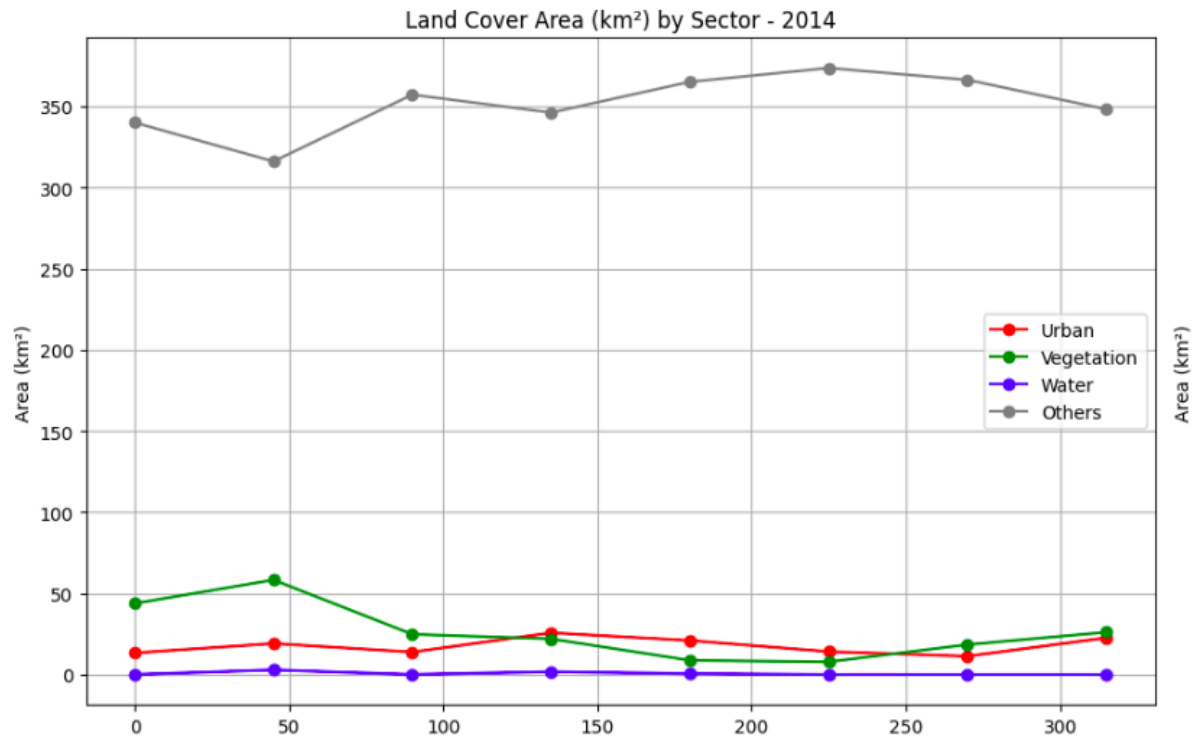
- **Percentage and area (in km²) of:**
 - Urban/Built-up
 - Vegetation
 - Water Bodies
 - Others

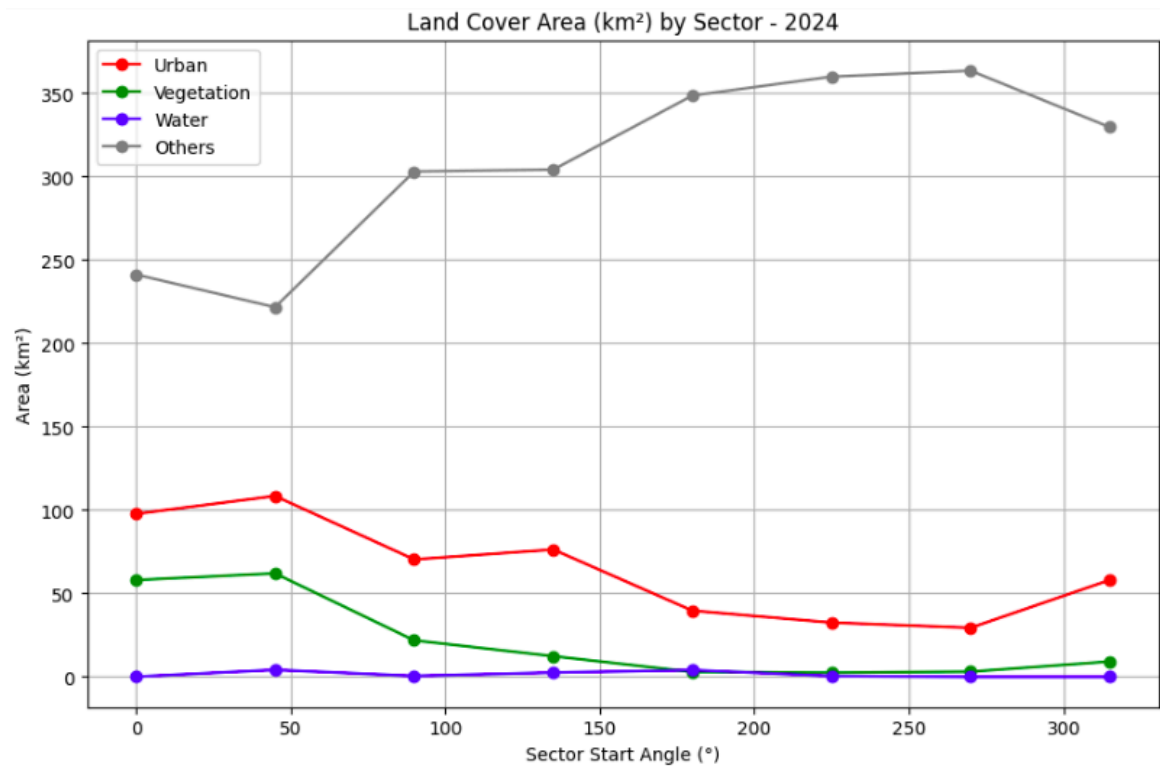
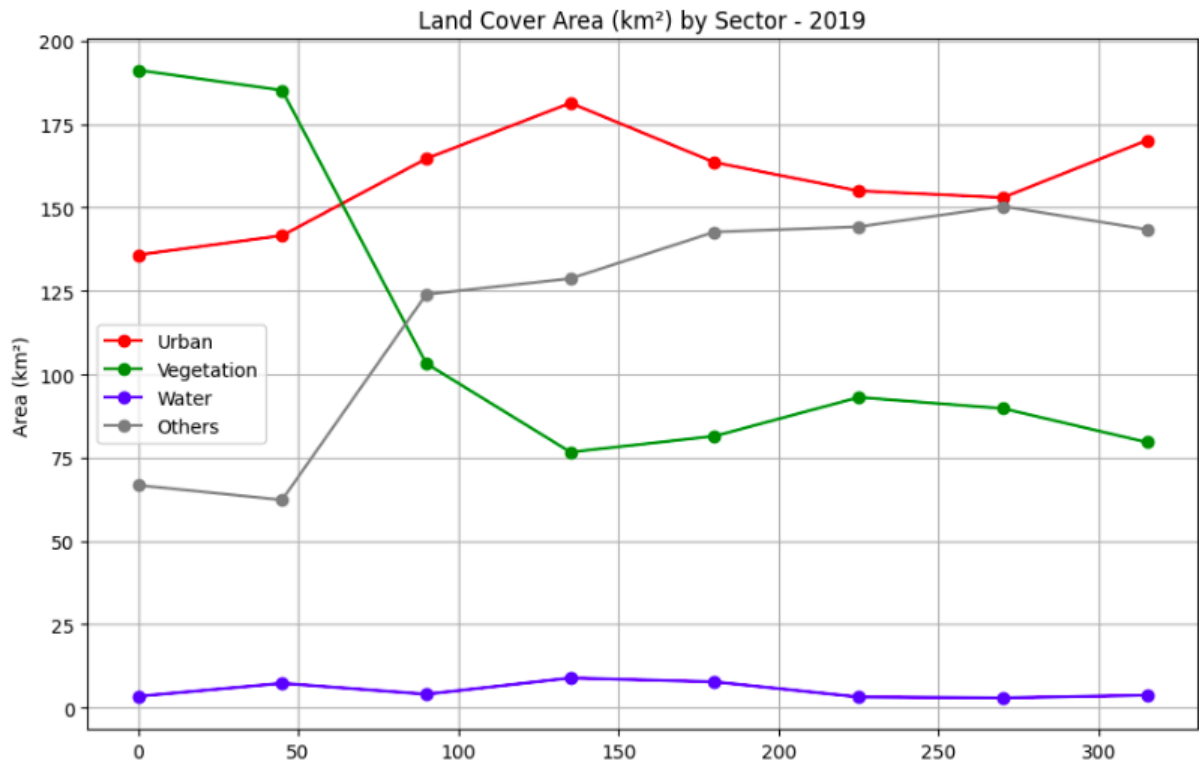
Table : Sector-wise land cover statistics from QDA classification

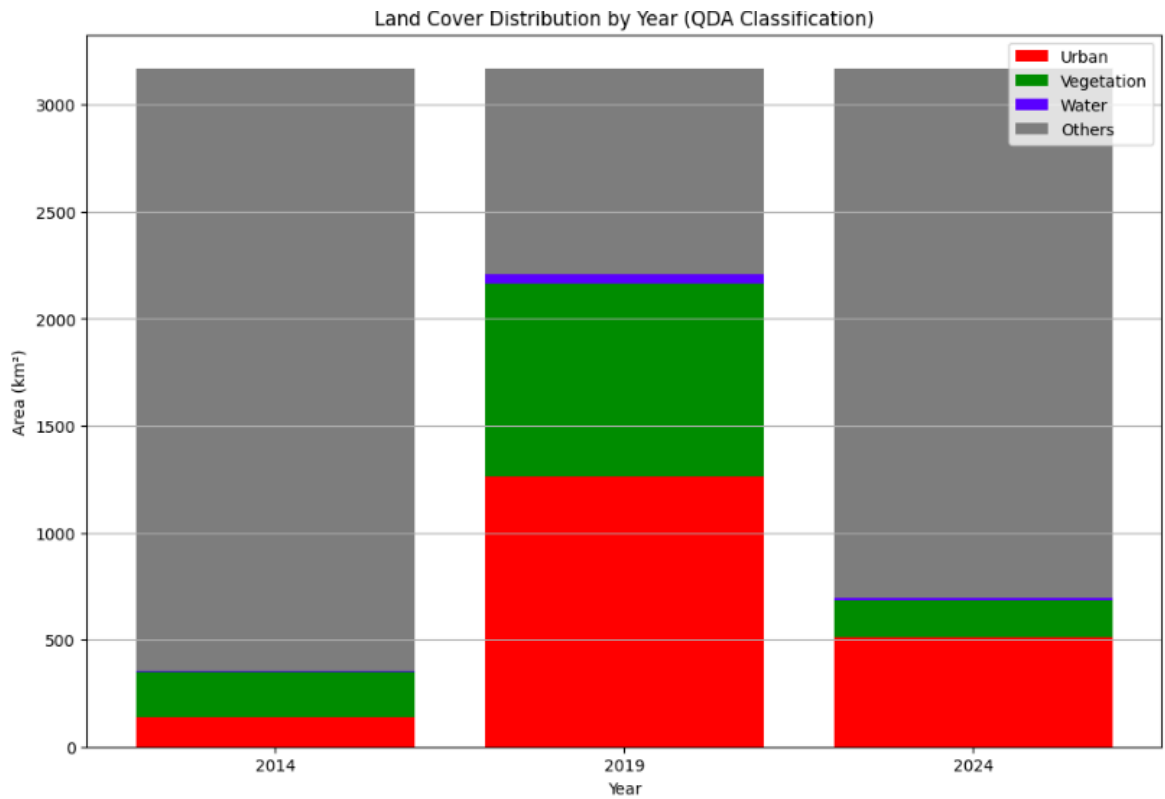
Year	Start Angle (°)	End Angle (°)	Urban %	Vegetation %	Water %	Others %	Urban (km ²)	Vegetation (km ²)	Water (km ²)	Others (km ²)	Total (km ²)
2014	0	45	3.36	11.03	0.01	85.6	13.33	43.81	0.05	339.93	397.12
2014	45	90	4.83	14.71	0.73	79.73	19.15	58.3	2.88	316.05	396.38
2014	90	135	3.5	6.27	0.01	90.22	13.86	24.82	0.05	357.13	395.86

2014	135	180	6.5	5.56	0.45	87.49	25.73	21.98	1.78	346.08	395.56
2014	180	225	5.31	2.22	0.14	92.33	20.99	8.77	0.57	365.04	395.37

Figure: Line plots showing Urban km² by sector over time







c) Sector-wise LST Extraction

- LST images for 2014, 2019, and 2024 were reduced over each sector polygon.
- Sector-wise **mean LST (in °C)** was calculated.

Table: Sector-wise Mean LST (°C) with corresponding land cover %

Year	Start Angle	End Angle	Mean LST (°C)	Urban (%)	Vegetation (%)	Water (%)	Others (%)
2014	0	45	28.73	66.16	33.6	0.01	0.23
2014	45	90	29.89	62.86	36.37	0.42	0.35
2014	90	135	32.26	76.51	23.11	0.03	0.35
2014	135	180	33.36	86.03	13.34	0.14	0.49
2014	180	225	34.87	90.96	8.85	0.05	0.14
2014	225	270	34.97	87.03	12.89	0	0.07
2014	270	315	29.3	79.85	20.01	0	0.13
2014	315	360	26.07	76.44	23.36	0	0.2
2019	0	45	39.81	41.29	58.43	0	0.28
2019	45	90	39.39	36.44	63.06	0.06	0.44
2019	90	135	42.81	50.36	49.15	0.1	0.39
2019	135	180	43.27	61.27	37.44	0.46	0.82

2019	180	225	43.02	55.88	43.45	0.15	0.52
2019	225	270	42.96	58.34	41.36	0.04	0.26
2019	270	315	43.5	59.17	40.66	0	0.17
2019	315	360	43.09	62.14	37.63	0	0.22
2024	0	45	32.37	61.86	37.9	0.01	0.23
2024	45	90	31.9	53.66	45.89	0.09	0.37
2024	90	135	34.58	71.77	27.77	0.11	0.35
2024	135	180	36.57	79.06	19.89	0.3	0.75
2024	180	225	38.14	83.89	15.05	0.7	0.36
2024	225	270	38.22	88.9	10.76	0.16	0.18
2024	270	315	35	89.95	9.94	0.01	0.1
2024	315	360	33.79	83.84	15.95	0.01	0.2

Figure: Scatter plot – Urban % vs LST per sector

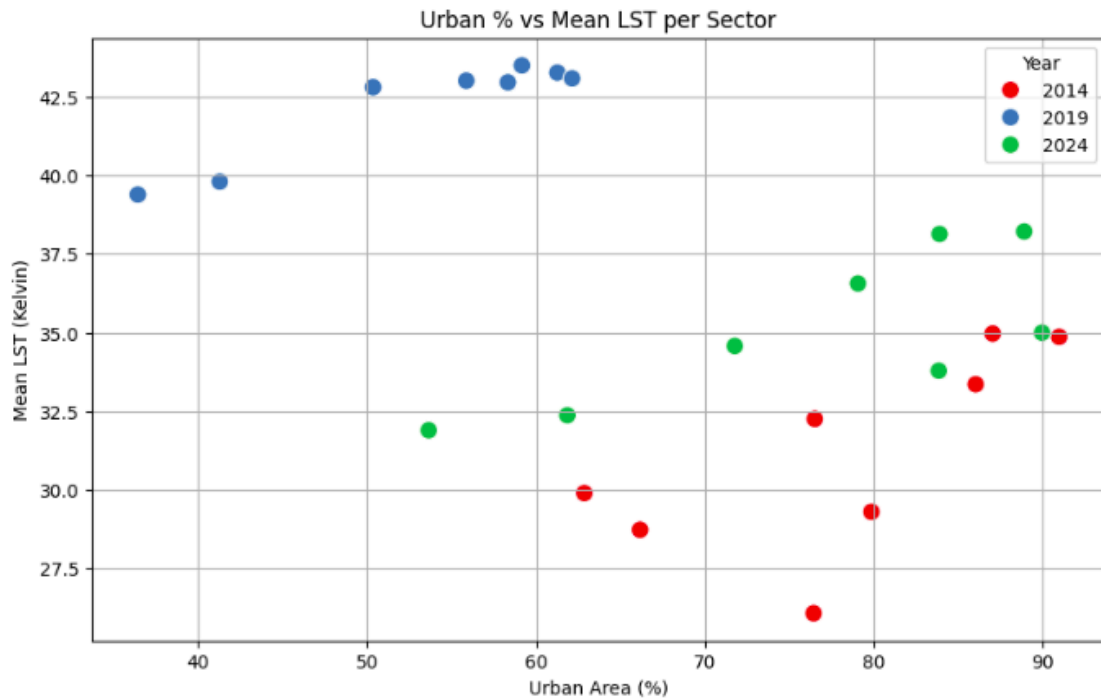
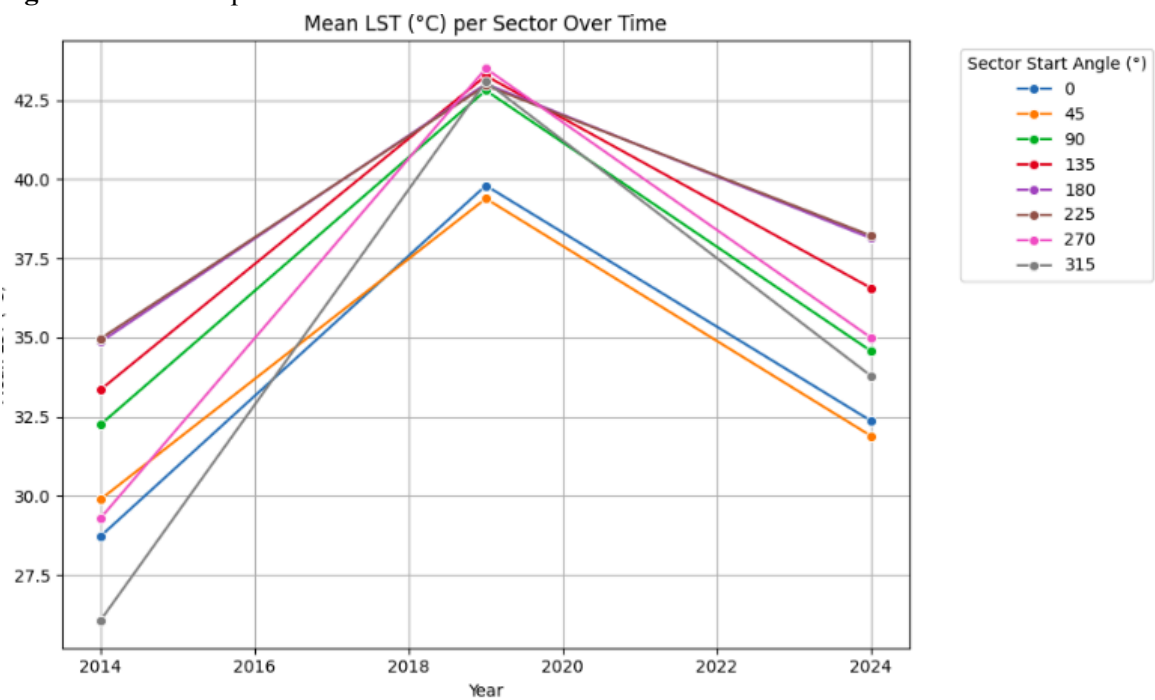


Figure: Mean LST per sector over time

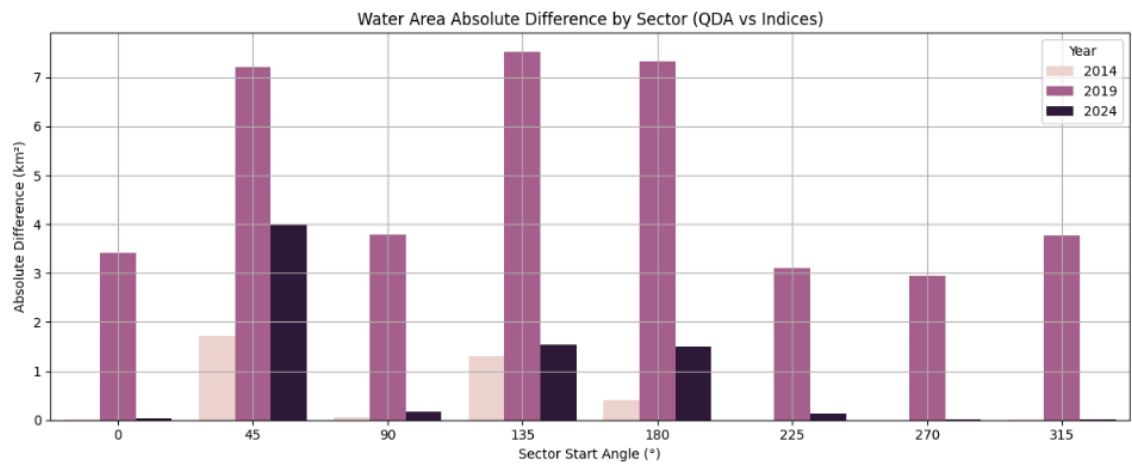
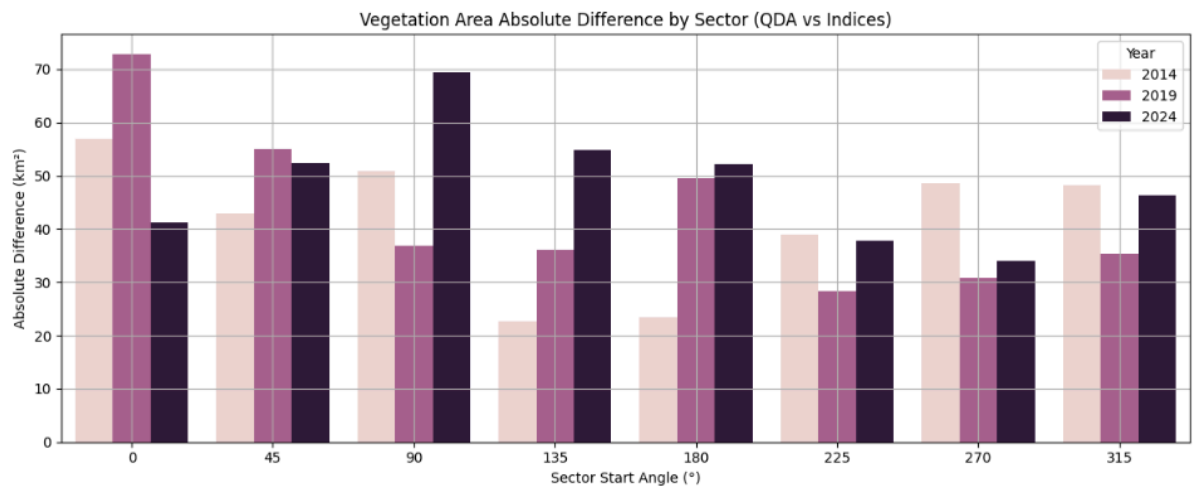
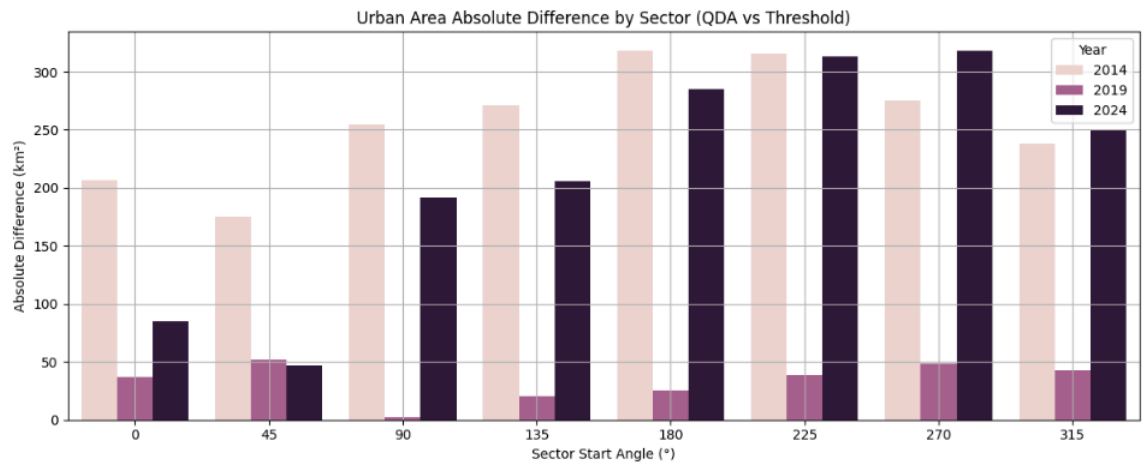


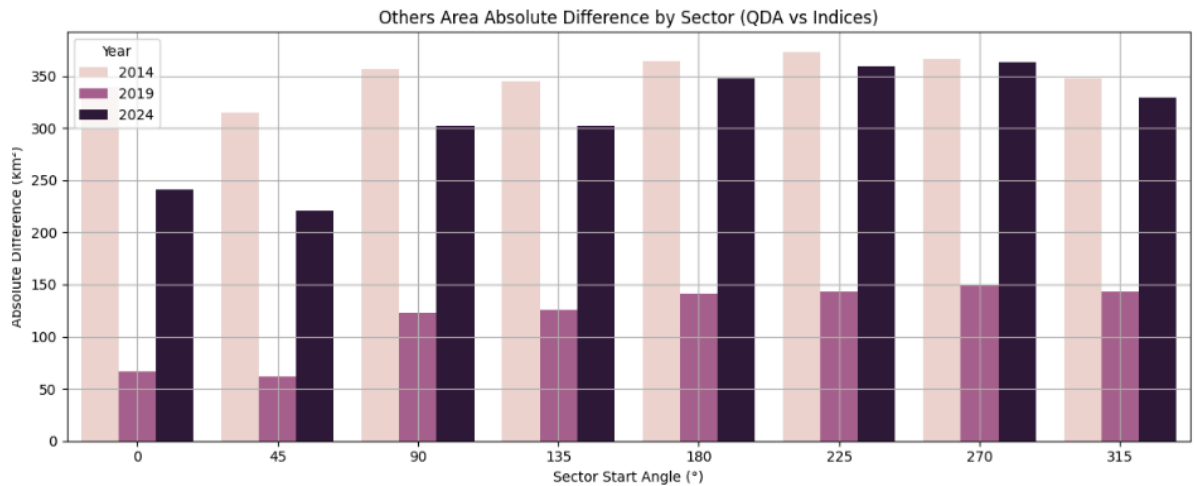
d) Comparative Analysis

A direct comparison was made between threshold-based and QDA classification outcomes.

Absolute differences (km²) in class-wise areas were plotted for each sector.

Figure: Class-wise absolute differences in area (km²) per sector





3.4.2 Water Body Buffer Analysis

a) Buffer Ring Construction

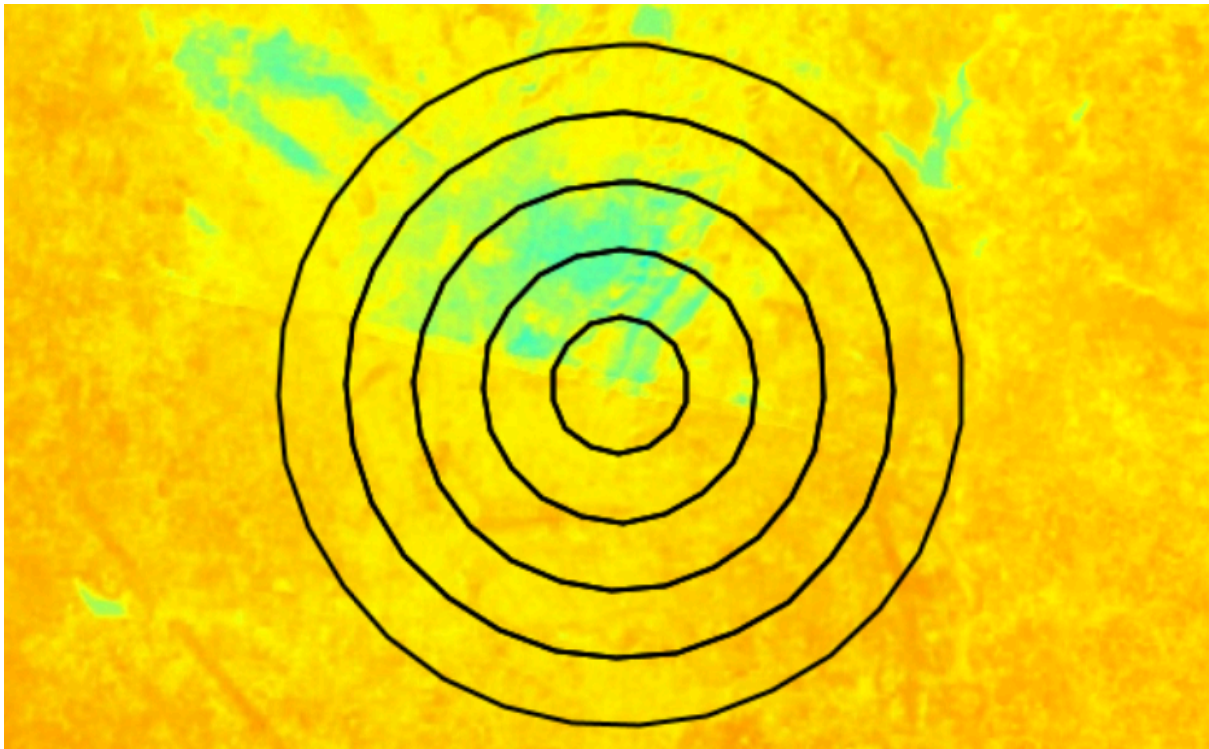
- **Man Sagar Lake** was selected as the focal water body.
- **Concentric rings of 2 km width** (from 2 km to 10 km) were generated around the lake using the `buffer()` and `difference()` functions in Earth Engine.

Figure: Buffer rings around Man Sagar Lake overlaid on LST map

b) LST Extraction by Ring

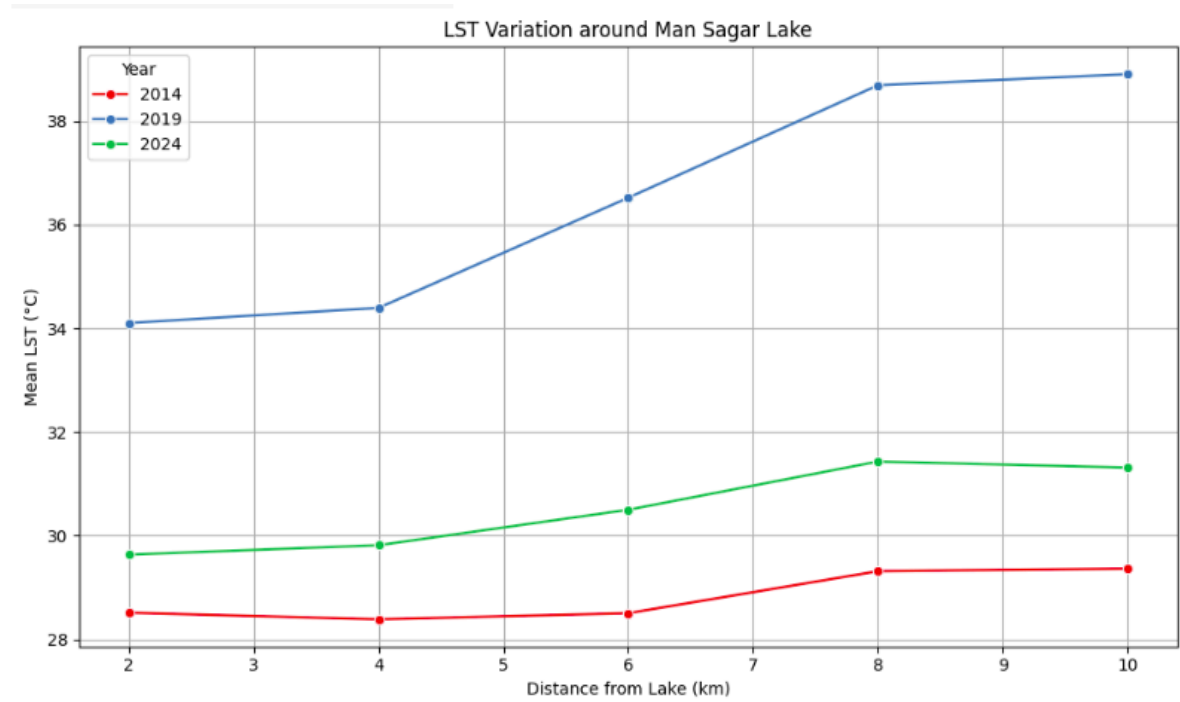
- For each buffer zone and each year (2014, 2019, 2024), **mean LST (°C)** was computed.
- This enabled tracking the **cooling gradient effect** of the lake across space and time.

Table: Mean LST (°C) by buffer distance and year



Distance (km)	Mean LST (K)	Year	Mean LST (°C)
2	301.664179	2014	28.514179
4	301.536348	2014	28.386348
6	301.655189	2014	28.505189
8	302.465473	2014	29.315473
10	302.516008	2014	29.366008
2	307.254796	2019	34.104796
4	307.547653	2019	34.397653
6	309.672933	2019	36.522933
8	311.842962	2019	38.692962
10	312.057285	2019	38.907285
2	302.785006	2024	29.635006
4	302.967007	2024	29.817007
6	303.650931	2024	30.500931
8	304.581684	2024	31.431684
10	304.465035	2024	31.315035

Figure: Line plot – LST vs Distance from Lake for 3 years



c) Interpretation of Lake Influence

- The cooling effect of the lake was strongest within 2–4 km in 2014, with LST gradually rising with distance.
- In 2019, urbanization around the lake reduced the cooling effect, seen as a sharper LST gradient.
- In 2024, some rebound is observed — potentially linked to water body restoration or seasonal changes.

3.5 Statistical Analysis

Statistical techniques were applied to quantitatively understand the relationships between land cover and land surface temperature (LST) at different spatial and temporal scales. Two core analyses were conducted:

1. **Correlation analysis** between land cover composition and LST
2. **Comparative analysis** of classification methods (Index-based vs QDA)

3.5.1 Correlation between Land Cover and LST

Pearson correlation coefficients were calculated between **mean LST (°C)** and the percentage of each land cover class (Urban, Vegetation, Water, Others) across 8 sectors per year (2014, 2019, 2024). Two classification approaches were used: **threshold-based** and **QDA (Quadratic Discriminant Analysis)**.

Table: Pearson Correlation – Threshold-Based Classification

Year	Urban % vs LST	Vegetation % vs LST	Water % vs LST	Others % vs LST
2014	0.68	-0.69	-0.02	-0.02
2019	0.95	-0.95	0.29	0.07
2024	0.76	-0.78	0.69	0.16

Table: Pearson Correlation – QDA-Based Classification

Year	Urban % vs LST	Vegetation % vs LST	Water % vs LST	Others % vs LST
2014	0.07	-0.56	0.09	0.49
2019	0.79	-0.98	-0.07	0.98
2024	-0.76	-0.82	0.13	0.79

Observations & Interpretation

- Urban % vs LST:**
 Strong **positive correlation** across all years using the threshold method, particularly in **2019** ($r = +0.95$), indicating that urbanized areas consistently contribute to increased surface temperatures.
 QDA results show more variability, with a **reversal in 2024**, suggesting potential classification artifacts or underlying spatial complexities.
- Vegetation % vs LST:**
 Shows a consistently **strong negative correlation** across both methods (e.g., -0.78 in 2024 threshold; -0.98 in 2019 QDA), reinforcing the cooling influence of vegetation.
- Water % vs LST:**
 Results are **mixed and inconsistent**, with weak correlations, possibly due to the small area share and variability in thermal behavior of water bodies.
- Others % vs LST:**
 Generally **weak to moderate positive correlation**, more pronounced in **QDA results** (e.g., +0.98 in 2019), possibly capturing mixed land types like barren or transitional surfaces.

Figure: Correlation between LST and Land Cover % by Year using Index based Classification

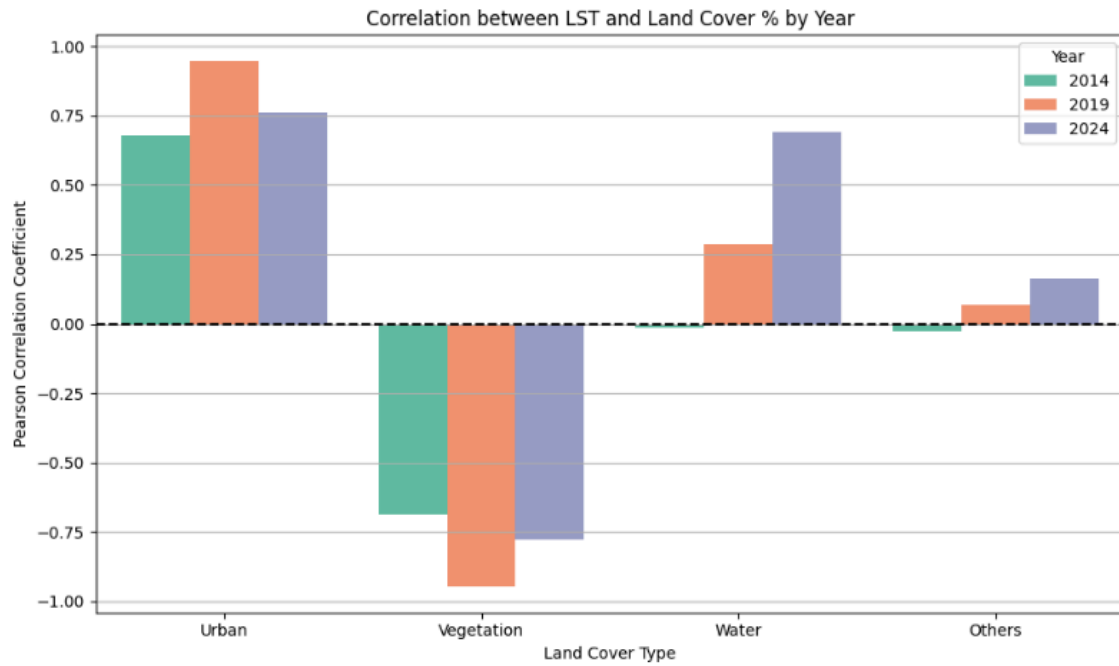
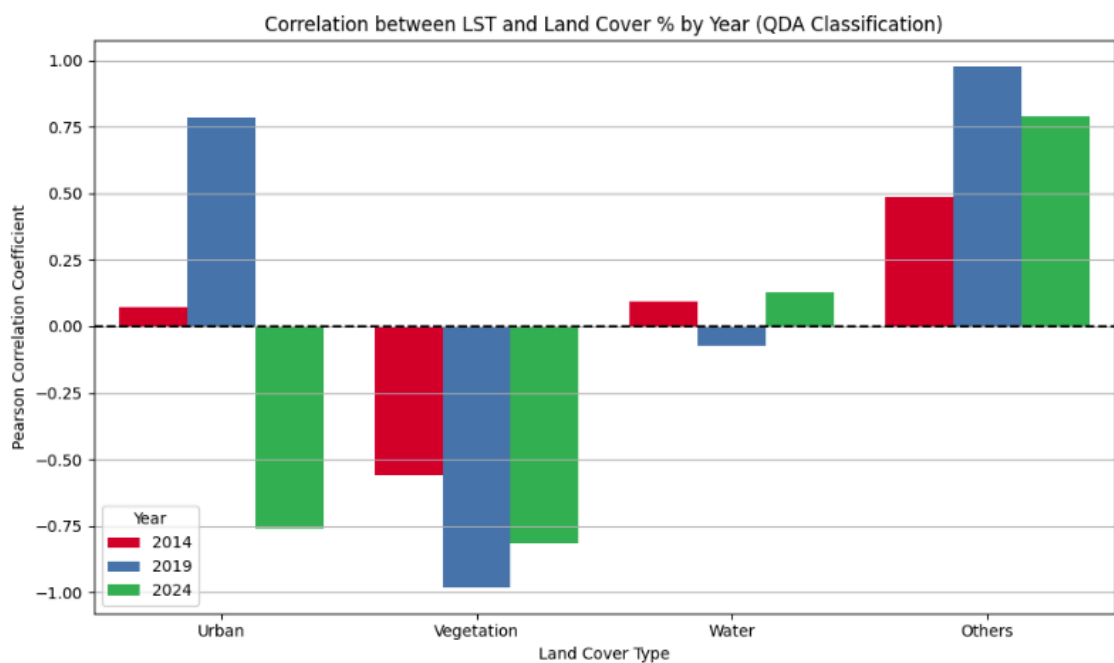


Figure: Correlation between LST and Land Cover % by Year using MLC



3.5.2 Comparative Analysis of Classification Methods

To evaluate the reliability of threshold-based and QDA-based land cover classification approaches:

- **Absolute area differences (km²)** were computed for each land cover class in each sector.
- QDA showed improved spatial detail and slightly different land cover estimates, especially in areas with mixed pixels or ambiguous index values.

Interpretation:

- Largest differences observed in **Urban** and **Others** classes.
- QDA performed better in heterogeneous zones, as seen from visual maps and model validation metrics.

3.5.3 Temporal Change Detection

Temporal trends from 2014 to 2024 revealed:

Urban Area

- Urban area increased **significantly across all sectors** between 2014 and 2024, with a notable **dip in 2019**, likely due to stricter cloud filtering or underestimation in classification.
- The **south (135°–225°) and southeast (225°–270°)** sectors showed the **steepest increases**, suggesting ongoing peri-urban expansion or sprawl in those directions.

Vegetation Area

- Vegetation peaked in **2019** in most sectors, potentially due to better post-monsoon greenness or temporary reforestation.
- From **2019 to 2024**, there was a **sharp decline in vegetation**, especially in the **south and southeast sectors**, aligning with the expansion of urban footprint and increased surface temperatures (LST).

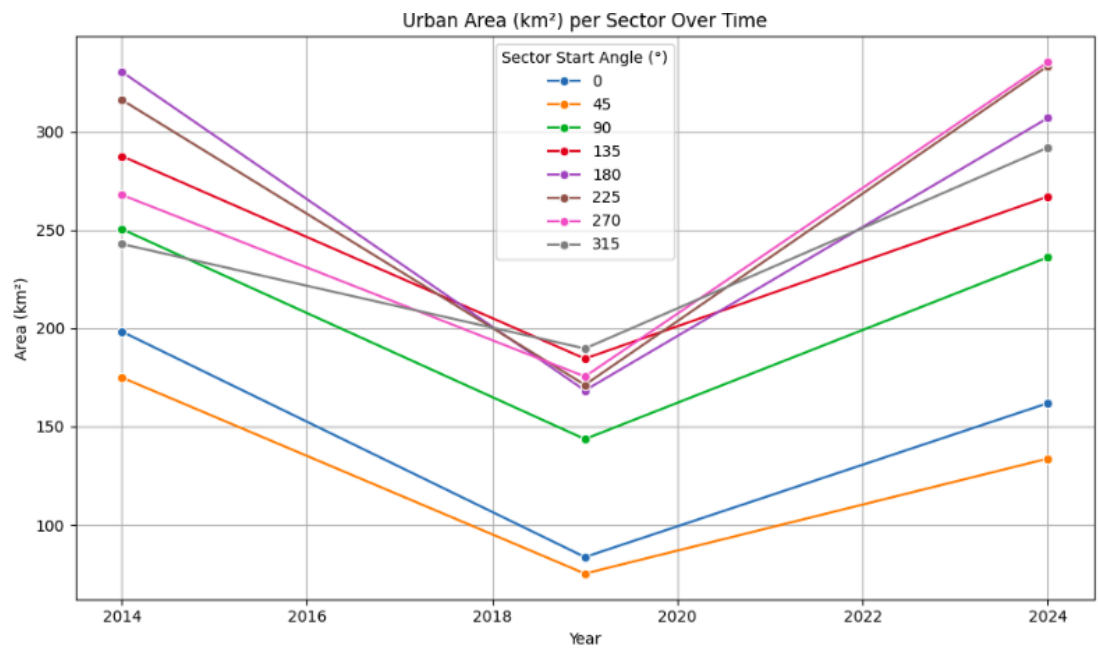
Water Area

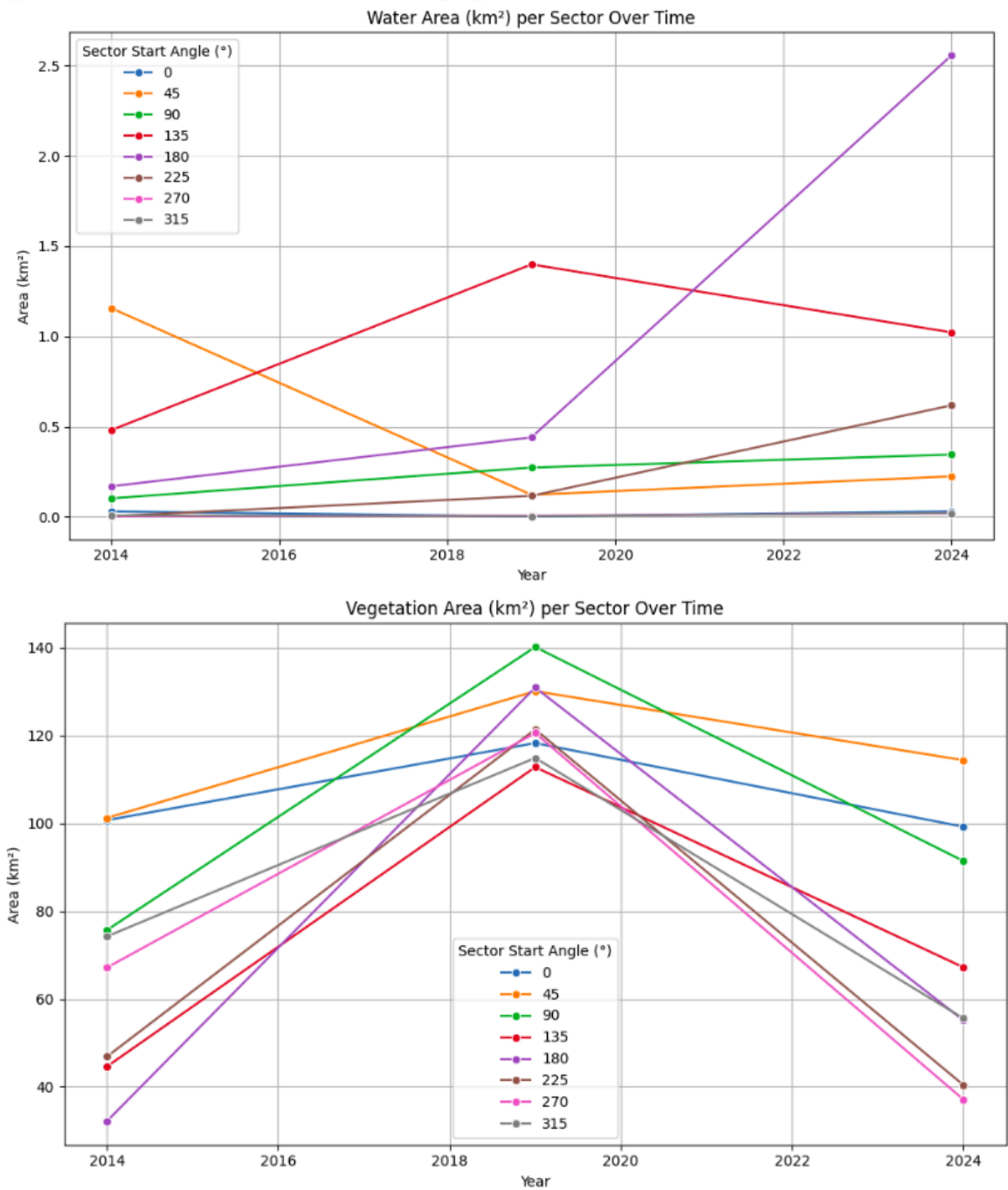
- Water coverage remained **minimal and highly variable**, with some sectors (notably **180°**) seeing an increase by 2024, while others (e.g., **45°**) experienced a sharp decline.
- These fluctuations may result from **seasonal surface water**, classification noise, or actual changes like new water bodies or drying of existing ones.

Others (Barren/Transition)

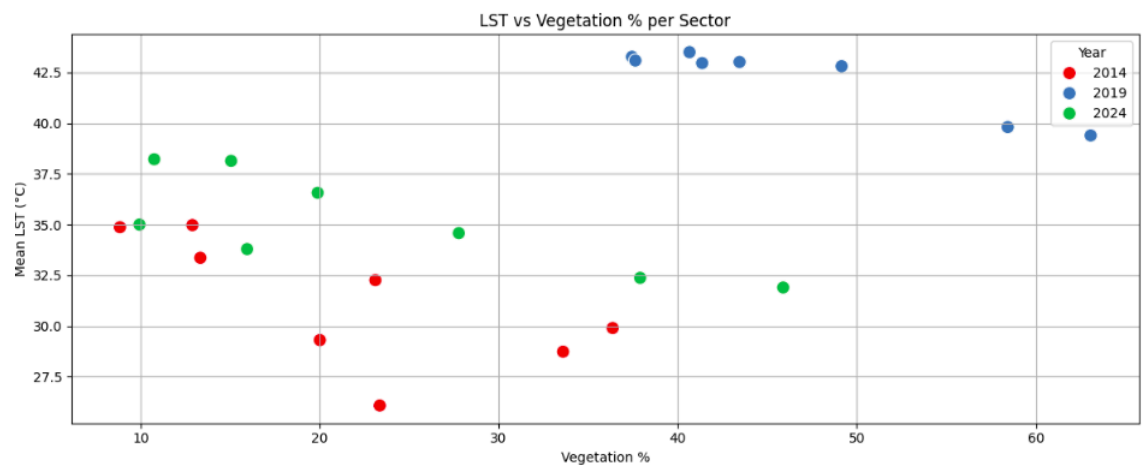
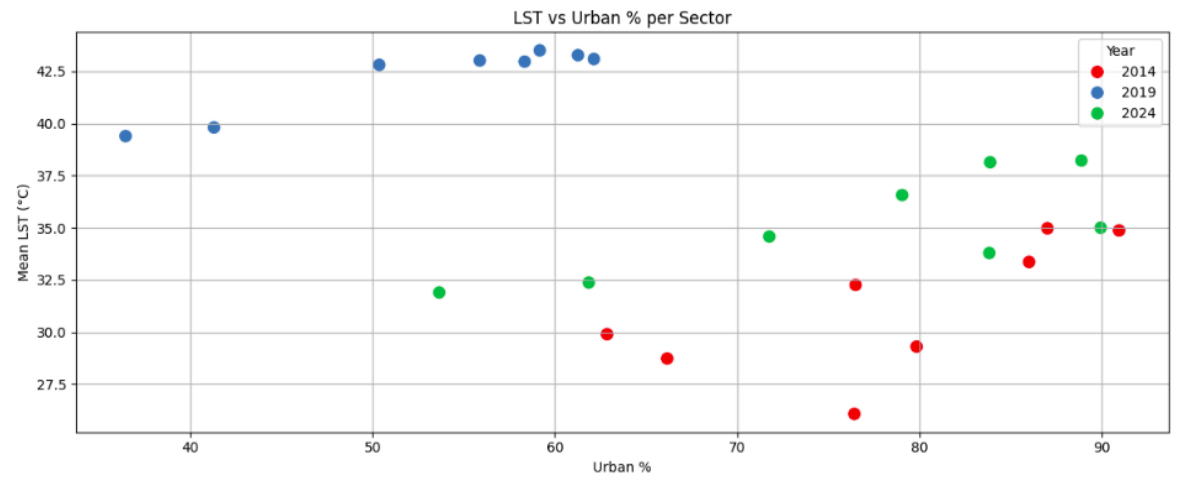
- The "Others" category showed a **general increase in 135° and 180° sectors**, suggesting more **bare soil or transitional land** due to construction or deforestation.
- Stability in other sectors indicates consistent land types or minimal disturbance.

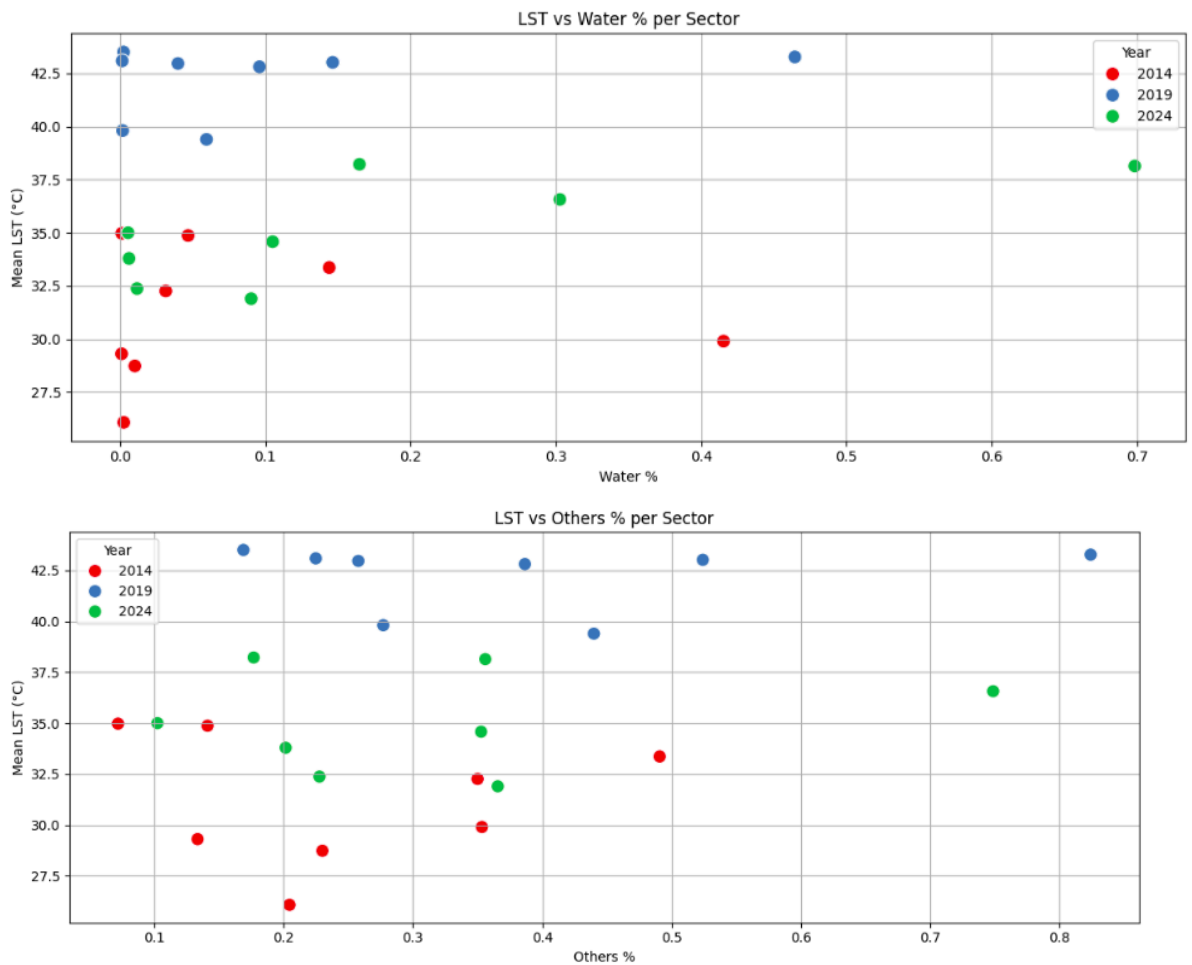
Figures: Land Cover change by year





- **LST increased steadily**, particularly from 2014 to 2019.
- Figures:** Change in Land Cover and LST over a decade for all sectors





4. Results and Discussion

4.1 Land Cover Classification Results

This section presents the results of land cover classification across the three time periods (2014, 2019, and 2024), using both threshold-based and machine learning approaches. It also evaluates classification accuracy, and compares performance between methods.

4.1.1 Spatial Distribution of Land Cover Types

Land cover classification maps show distinct spatial patterns of urbanization, vegetation concentration, and water body extent. Urban and built-up areas are clustered primarily in the central and southeastern sectors, while vegetation is denser in peripheral and northern regions.

Observations:

- **Urban expansion** is clearly visible from 2014 to 2024, especially toward the east and south of the city but decreased little in 2019.
- **Vegetation cover** shows fragmentation and reduction, particularly in the core sectors.

- **Water bodies** (e.g., Man Sagar Lake) remain relatively stable but fluctuate slightly in classification results.
- **‘Others’ class** mostly aligns with barren land or unclassified mixed pixels at the urban fringe.

4.1.2 Classification Accuracy Assessment

Model evaluation metrics demonstrate that both **QDA and Random Forest** performed well, with Random Forest achieving perfect or near-perfect accuracy across all years.

Notes:

- QDA showed strong but slightly variable performance.
- RF was more robust, particularly in distinguishing vegetation and others.
- Minor misclassifications in QDA occurred between Urban/Others and Vegetation/Others in 2024.

4.1.3 Comparison of Classification Methods

Comparing threshold-based and QDA classifications highlights key differences in boundary definitions and spatial consistency.

Observations:

- **Urban areas** tend to be slightly underclassified in threshold-based methods due to reliance on a fixed NDBI threshold.
- **QDA captures mixed-use zones** and edge conditions better (e.g., industrial parks).
- **Vegetation is overestimated** in threshold method in sectors with interspersed fallow land.
- The **Others class** shows the highest disagreement, due to its broad residual definition.

Interpretation:

- QDA provided improved discrimination in complex zones, consistent with its multivariate logic.
- However, threshold-based classification is computationally simpler and reproducible and accurate to a extent.

4.2 Temporal Land Cover Changes & Intra-Year LULC (2014–2024)

This section presents the quantitative and spatial dynamics of land cover transition in Jaipur over the study period, focusing on both overall and sectoral changes. The analysis is based primarily on **QDA classification results**, which were found to be more spatially robust and accurate.

4.2.1 Overall Land Cover Transition

Land cover area for each class (Urban, Vegetation, Water, Others) was summed over the region of interest for 2014, 2019, and 2024 using QDA outputs.

Observations:

- **Urban area** increased significantly from 2014 to 2024.
- **Vegetation cover** showed a moderate decline.
- **Others class** remained dominant in total area but slightly decreased.
- **Water bodies** fluctuated only marginally.

Interpretation:

- Urban expansion occurred at the cost of both vegetation and unclassified (Others) areas.
- Water body stability suggests limited encroachment, possibly due to topography and regulation.

4.2.2 Sectoral Land Cover Dynamics

To understand the directional trends of change, the study area was divided into **eight 45-degree sectors** from the city center. Area statistics for each class were computed per sector and per year.

Observations:

- **Southeastern (135°–180°)** and **Southern (180°–225°)** sectors experienced the most aggressive **urban expansion**.
- **Vegetation declined** particularly in the **Northern (0°–45°)** and **Western (270°–315°)** sectors in 2024.
- **Water coverage** was largely stable across sectors (as expected, mostly constrained to the lake buffer).

Interpretation:

- Sectoral patterns highlight the **directional nature** of Jaipur’s urban sprawl.
- Land cover changes are not uniform — spatial analysis identifies **priority zones** for policy focus (e.g., urban containment, green space conservation).

4.3 Land Surface Temperature Patterns

LST was derived using the **Mono-window algorithm** applied to Landsat 8 thermal bands for the dry season (Oct–Dec) in 2014, 2019, and 2024. The results reveal both spatial heterogeneity and temporal escalation in surface temperatures, strongly tied to land cover characteristics.

4.3.1 Spatial Distribution of LST

LST maps highlight the **thermal structure** of the study region at each time step.

Observations:

- Urban centers consistently exhibit **higher temperatures** (thermal hotspots).
- **Cool zones** align with vegetation-rich areas, open fields, and water bodies (e.g., Man Sagar Lake).
- In 2019, heat concentration expanded radially, indicating intensification of the **urban heat island (UHI)** effect.

Interpretation:

- Spatial LST patterns reflect the **synergy between land use and surface temperature**, with built-up zones acting as heat sinks.

4.3.2 LST Variations

Mean LST values were extracted for each sector and compared across years.

Observations:

- **Average LST increased** across almost all sectors between 2014 and 2019.
- Slight **cooling or stabilization** occurred in 2024, potentially due to seasonal differences or reforestation in some sectors.
- **Highest LST in 2019**, especially in central and south-eastern zones.

Interpretation:

- The LST spike in 2019 may result from **climatic anomalies** (hotter dry season), **peak urban activity**, or reduced vegetation cover.
- These trends underline the **importance of year-on-year monitoring** for long-term urban climate strategies.

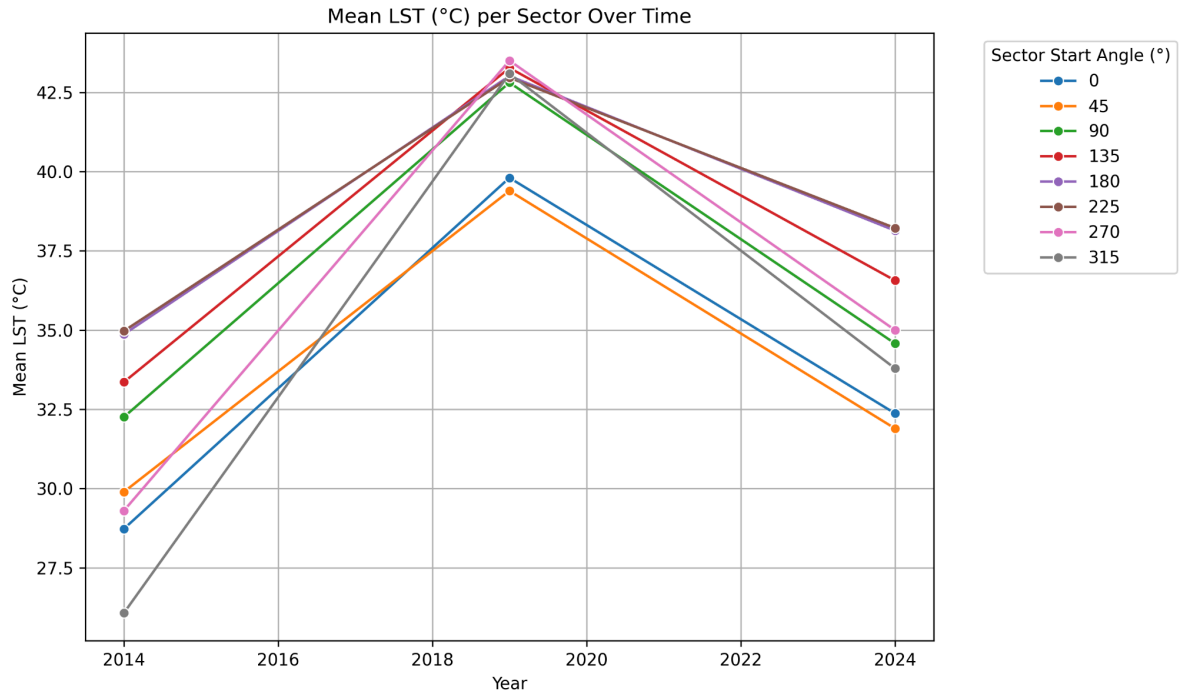


Fig.

4.4 Relationship Between Land Cover and LST

This section explores the statistical and spatial connections between land cover categories and Land Surface Temperature, both sectorally and temporally. Two classification methods (Threshold-based and QDA) were used for comparative correlation analysis.

4.4.1 Correlation Analysis

Pearson correlation coefficients were calculated between **mean LST** and percentage cover of each land class (Urban, Vegetation, Water, Others) for every year and classification method.

Observations (Threshold-based):

- **Urban_ %**: Strong positive correlation with LST ($r \approx 0.68$)
- **Vegetation_ %**: Strong negative correlation with LST ($r \approx -0.69$)
- **Water_ %**: Weak to moderate correlation, inconsistent
- **Others_ %**: Minimal to weak positive correlation

Observations (QDA-based):

- **Vegetation_ %**: Correlation strengthened to -0.56 in 2019
- **Urban_ %**: Correlation was close to 0.07

- **Others_ %**: Showed strong positive correlation in 2019, suggesting inclusion of built-up-like areas

Interpretation:

- Vegetation consistently exerts a **cooling effect**.
- QDA classification captures more nuanced spatial patterns, affecting correlation structure.
- “Others” is a **heterogeneous class**, and its correlation varies depending on how it spatially overlaps with urban or barren surfaces.

4.4.2 Urban Areas and LST

Observations:

- Sectors with >50% urban area consistently exhibit **higher mean LST**.
- 2019 shows the sharpest gradient in LST rise with increasing urbanization.

Insight:

- The urban heat island effect is not only persistent, but also **intensifies in sectors with dense development** and limited green cover.

4.4.3 Vegetation and LST

Observations:

- Strong inverse relationship, especially in 2019 and 2024.
- Even 20–30% vegetation cover within a sector can **lower LST**.

Interpretation:

- Confirms the **mitigating role of green cover**.
- Suggests critical thresholds of vegetation cover needed to curb extreme temperatures.

4.4.4 Water Bodies and LST

This subsection focuses on **Man Sagar Lake**, using 2 km buffer rings to analyze thermal gradients outward from the lake.

Observations:

- In 2019, LST increases from ~34.1°C at 2 km to ~38.9°C at 10 km. Again near water bodies/vegetation it is little lesser.
- Cooling effect is there from the lake .

Interpretation:

- Water bodies act as **climatic buffers**, but their effect **diminishes beyond 6–8 km**.
- Cooling efficiency may vary with **seasonal evaporation rates** and **lake size stability**.

4.4.5 “Others” Land Class and LST

Observations:

- Correlation with LST is inconsistent — positive in some years, weak in others.
- Likely includes **bare soil, wasteland, transitional zones**, each with varied thermal response.

Interpretation:

- “Others” is a **residual class**; its behavior depends on contextual composition.
- May mask or dilute precise analysis, underscoring the importance of finer subclassification.

4.5 Man Sagar Lake: Land Cover and Thermal Analysis

Man Sagar Lake, located in the northeastern quadrant of Jaipur, plays a role in the city’s urban ecology and microclimate. Its spatial and thermal characteristics were analyzed separately to assess its behavior across the study period.

Land Cover Observations

- The **lake’s spatial footprint remained relatively slightly stable** from 2014 to 2024, with only minor fluctuations in classified area due to seasonal water level variation and classification differences.
- In some years (e.g., **2019**), the lake area appeared slightly larger, possibly due to post-monsoon water retention or clearer satellite imagery.
- **Surrounding land** was consistently classified under “**Others**” or **Vegetation**, reflecting **barren or mixed-use fringe** areas.
- Dryness is increasing at a **slow** pace since 2014.

Thermal (LST) Observations

- Man Sagar Lake consistently appeared as a **cool zone** in all LST maps, with significantly lower surface temperatures than adjacent urban or barren areas.
- A buffer analysis using concentric rings (2 km to 10 km) revealed a **pronounced cooling gradient**, especially in **2019**, when temperatures rose from **~34.1°C (2 km)** to **~38.9°C (10 km)**.
- The **cooling effect was strongest within a 2–4 km radius**, beyond which the influence diminished due to built-up encroachment and land use transitions.

Interpretation

- The lake functions as a **localized thermal moderator**, with a measurable impact on surrounding land surface temperatures.
- Its **stability over time** reflects effective natural buffering or protected status, but **urban pressure around the lake** remains a concern.

- **Preserving and enhancing the buffer zone** around Man Sagar Lake could help **amplify its cooling potential**, especially in the face of rising urban heat.

4.6 Implications for Urban Planning and Climate Adaptation

The integration of land cover classification and LST analysis across a decade reveals critical insights for urban sustainability, thermal comfort, and climate adaptation in Jaipur.

4.6.1 Evidence-Based Cooling Strategies

- The **strong negative correlation** between vegetation and LST across all years confirms the **cooling potential of green infrastructure**.
- Spatial patterns reveal that **sectors with higher green cover consistently recorded 2–4°C lower LST** than heavily urbanized zones.

Recommendation:

Implement **urban greening strategies** such as green belts, vertical gardens, and rooftop vegetation especially in sectors 180°–270°, which showed the highest LST increases.

4.6.2 Priority Intervention Zones

Using overlay analysis of land cover and thermal maps, the following zones emerge as **priority areas**:

- **Southeastern (135°–225°) and Southern (180°–270°)** sectors showed:
 - Highest urban growth
 - Most significant LST rise
 - Lowest vegetation retention

Recommendation:

Urban planning in these sectors should **mandate green buffers, preserve open land**, and avoid dense construction near existing heat-prone zones.

4.6.3 Land Use Modifications with Cooling Impact

The results demonstrate that **urban expansion without vegetation buffers** exacerbates the UHI effect. Conversely, proximity to **water bodies and vegetated zones** helps lower surface temperatures.

Suggested Interventions:

- **Mixed-use planning** that incorporates permeable surfaces and vegetation corridors
- **Zoning laws** to cap impervious surface ratio per plot
- Preservation and expansion of **urban wetlands** and lakeside buffers

4.6.4 Long-Term Adaptation Vision

The temporal analysis across 2014–2024 shows:

- A consistent **urbanization trend**, with ~60% increase in built-up area
- A **sharp rise in LST** by up to 10°C in urban sectors
- **Vegetation loss** coinciding with these trends

Planning Implications:

- Data-driven land use planning is **no longer optional**, especially in climate-vulnerable zones like Jaipur.
- This study supports a **GIS-integrated planning framework**, where LST and land cover dynamics are continually monitored to inform:
 - **Building codes**
 - **Infrastructure placement**
 - **Green cover targets**

5. Conclusion

This study investigated the spatiotemporal dynamics of land cover and their relationship with Land Surface Temperature (LST) in Jaipur over a decade (2014–2024), using both index-based and machine learning classification techniques integrated with satellite data analysis. The results offer critical insights into urban expansion, surface heating patterns, and the thermal moderation role of green and blue spaces.

5.1 Key Findings

Land Cover Change

- Between 2014 and 2024, built-up area increased significantly, especially in the **southern and southeastern sectors** (135°–270°).
- Vegetation cover declined in most sectors, though some recovery was seen in 2024 compared to 2019.
- Water bodies remained spatially limited, with marginal changes.

LST Patterns

- Mean LST values increased across most sectors from **2014 to 2019**, with the **peak heat load observed in 2019**.
- By 2024, there was a modest decline in LST in some sectors, potentially due to vegetation rebound or climatic variation.

Land Cover–LST Relationships

- Strong **positive correlation** between Urban_% and LST in threshold-based classification.
- **Negative correlation** between Vegetation_% and LST, consistent across all years and methods.

- QDA-based classification revealed **more complex patterns**, particularly in the “Others” class.

Water Body Cooling Effect

- Analysis of buffer rings around Man Sagar Lake revealed a **cooling gradient** from 2 km to 10 km in 2019.
- The effect was **most pronounced during peak heating years**, affirming the role of lakes as **natural thermal regulators**.

5.2 Implications

Urban Planning in Jaipur

- Results underscore the **urgent need to preserve and expand vegetation** within and around urban growth corridors.
- Sector-wise analysis enables **targeted intervention planning**, helping prioritize high-risk heat zones.

Applications in Other Cities

- The integrated approach used here is adaptable to other **semi-arid or rapidly urbanizing cities**.
- LST–land cover correlations offer a template for **green infrastructure prioritization** in urban master plans.

Methodological Contributions

- Demonstrated the **strengths and trade-offs** between index-based and machine learning classification approaches.
- Showed how remote sensing, geospatial statistics, and classification models can be combined for **urban thermal analysis**.

5.3 Limitations and Future Research Directions

Limitations

- **Training sample size** was limited, especially in marginal classes like water or others.
- **Single-season analysis** may not fully capture intra-annual LST or vegetation variation.
- LST estimates are based on **surface temperature**, not air temperature, and may differ in magnitude.

Future Research

- Incorporate **multiseasonal or multi-year composites** to assess longer-term trends.
- Expand training datasets using **citizen science or UAV-based sampling**.
- Apply **object-based or deep learning classifiers** for better distinction of complex land covers.

- Integrate **socioeconomic data** to assess urban heat exposure risks by population vulnerability.

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- The open-source communities behind **Colab**, **Python**, **geemap**, **scikit-learn**, **matplotlib**, **seaborn**, and **rasterio**, whose libraries were instrumental in data analysis, machine learning, and cartographic rendering. All our implementation has been done in Colab. Corresponding output files are visible on execution of code and in the submitted data files.
- Our academic mentors and institutional guides for their consistent feedback, domain insights, and critical review throughout the course of this study.

This work was carried out as part of a final academic evaluation project, and the guidance received has been invaluable in shaping its direction and depth.

Appendix

Link to Colab Notebook: [Link](#)

Confusion matrices for 2019 and 2024

